ALPHA MICROSYSTEMS
RIGHT. FROM THE START.

AlphaCALC®
Tutorial

DSM-00166-00

© 1995 Alpha Microsystems

REVISIONS INCORPORATED	
REVISION	DATE
A00	October 1991

AlphaCALC Tutorial.

To re-order this document, request part number DSO-00166-00.

The information contained in this manual is believed to be accurate and reliable. However, no responsibility for the accuracy, completeness or use of this information is assumed by Alpha Microsystems.

This document may contain references to products covered under U.S. Patent Number 4,530,048.

The following are registered trademarks of Alpha Microsystems, Santa Ana, CA 92799:

AMIGOS	AMOS	Alpha Micro	AlphaACCOUNTING
AlphaBASIC	AlphaCALC	AlphaCOBOL	AlphaDDE
AlphaFORTRAN 77	AlphaLAN	AlphaLEDGER	AlphaMAIL
AlphaMATE	AlphaNET	AlphaPASCAL	AlphaRJE
AlphaWRITE	CASELODE	OmniBASIC	VER-A-TEL
VIDEOTRAX			

The following are trademarks of Alpha Microsystems, Santa Ana, CA 92799:

AlphaBASIC PLUS	AlphaVUE	AM-PC	AMTEC
AlphaDDE	AlphaConnect	DART	*in*Sight/am
*in*Front/am	ESP	MULTI	

All other copyrights and trademarks are the property of their respective holders.

ALPHA MICROSYSTEMS
2722 S. Fairview St.
P.O. Box 25059
Santa Ana, CA 92799

TABLE OF CONTENTS

PREFACE.....iii

INTRODUCTION
What is AlphaCALC?.....I-1
AlphaCALC Worksheets.....I-1
AlphaCALC'S Design.....I-2
What This Book Does.....I-3
Who This Book is for.....I-3
How to Use This Book.....I-4
The Structure of the Tutorial.....I-4
How the Lessons Are Organized.....I-5
Page Format and Conventions.....I-5
A Note on Screen Illustrations.....I-7
Some Notes on Keys.....I-7
About the Sample Worksheet.....I-8

LESSON ONE - GETTING STARTED
1 – Calling Up AlphaCALC.....1-1
2 – Viewing Help Information.....1-2
3 – Creating a Worksheet.....1-4
4 – The Worksheet Header.....1-6
5 – The Worksheet Display and the Menu Key.....1-9
6 – Storing Your Worksheet.....1-2
7 – Leaving AlphaCALC.....1-3

LESSON TWO - ENTERING DATA
1 – Calling Up Your Worksheet.....2-1
2 – Moving Around Your Worksheet.....2-4
3 – The Help Key.....2-5
4 – Entering Numbers.....2-7
5 – Entering Text.....2-9
6 – Speeding Up Entry.....2-11
7 – Math Calculations.....2-13
8 – More Practice.....2-16
9 – The SUM Function.....2-17

LESSON THREE - PRINTING YOUR WORKSHEET
1 – The Main Menu Print Command..... 3-1
2 – Choosing a Worksheet and Options..... 3-2
3 – Selecting a Printer..... 3-4

LESSON FOUR - ADVANCED DATA ENTRY
1 – Moving Your Worksheet Display..... 4-1
2 – Changing a Cell Formula..... 4-4
3 – Recomputing Your Worksheet..... 4-5
4 – Entering a Percentage..... 4-6
5 – Setting Cell Formats..... 4-9
6 – Setting Cell Alignment..... 4-13
7 – The IF Function..... 4-17
8 – The Save Command..... 4-19

LESSON FIVE - REARRANGING YOUR WORKSHEET
1 – Inserting and Deleting Rows..... 5-1
2 – Marking a Block of Cells..... 5-4
3 – Copying a Block of Cells..... 5-6
4 – Clearing Block Marks..... 5-8
5 – More Copying..... 5-9
6 – Formula Adjustment..... 5-11
7 – Storing Your Worksheet..... 5-12
8 – The Global Command..... 5-13
9 – The Editing Menu Quit Command..... 5-16
A Last Word.....5-18

APPENDIX A - FOR MORE PRACTICE
1 – The Year to Date Amounts..... A-1
2 – The Planned Amounts..... A-3
3 – Comparing Planned and Actual..... A-4

GLOSSARY

DOCUMENT HISTORY

INDEX

READER'S COMMENTS

PREFACE
THE AlphaCALC DOCUMENTATION LIBRARY

The *AlphaCALC Tutorial* is just one part of the AlphaCALC Documentation Library. Each part of the library is designed for a separate, specific purpose, so you can always get just the kind of information you need. The books in the AlphaCALC Documentation Library are:

- The *Release Notes* for your version of AlphaCALC contain compatibility information and general descriptions of new and changed features.

- *The AlphaCALC Tutorial*. This volume is your introduction to AlphaCALC. Starting with how to call up AlphaCALC and create a worksheet, it leads you step-by-step through all you need to know to begin using AlphaCALC.

- *The AlphaCALC Reference Book*. This document is your everyday guide to AlphaCALC. It has sections on AlphaCALC commands, key features, and functions, and also describes general procedures and topics of interest throughout AlphaCALC. If you want to know how to use the **SORT** command, the proper format of the IF function, what cell formats you can use, or just about anything else about AlphaCALC, this is the place to look.

- *The AlphaCALC Installation and Administrator's Guide*. This document tells you how to install AlphaCALC. It also covers topics of interest to the system administrator, including the AlphaCALC initialization file, graph setup, memory requirements, and how to change AlphaCALC screens.

- *AlphaCALC Programmer's Information*. This document contains information about the more technical features of AlphaCALC, including: the interface to AlphaBASIC and assembly language programs; DIF and other import/export file formats; technical functions, such as those dealing with ASCII values and octal and hexadecimal numbers; the different types of numeric data AlphaCALC supports, and more.

- *The AlphaCALC Keyboard Command Quick Reference Card*. This handy reference tool lists how to call up all of AlphaCALC's key features for all terminals Alpha Micro supplies.

You can order all these documents from your Alpha Micro dealer.

INTRODUCTION

The *AlphaCALC Tutorial* is your introduction to the world of AlphaCALC®. Even if you've never used any electronic spreadsheet, you can start using AlphaCALC with this tutorial. Starting with how to call AlphaCALC up on your terminal screen for the first time, you learn all you need to know to let AlphaCALC make your job easier.

WHAT IS AlphaCALC?

AlphaCALC is Alpha Micro's electronic spreadsheet system. Like all electronic spreadsheets, AlphaCALC can make your work easier by making it simpler and faster for you to do the calculations you need. The next sections discuss the two basic aspects of AlphaCALC: how it organizes information, and how its design makes it easy to use.

AlphaCALC WORKSHEETS

In AlphaCALC, we call the spreadsheet a "worksheet," since it allows you to work with information. A worksheet is a collection of "cells," arranged in rows, columns, and layers, like a pad of graph paper or checkerboards stacked on top of each other. You put your data—numbers, titles, and so on—in some of these cells, and the calculations you want AlphaCALC to perform in others.

Each row of the worksheet is numbered, and each column and each layer is lettered. For example, this is what a checkerboard would look like as one layer of an AlphaCALC worksheet:

	A	B	C	D	E	F	G	H
1	A1	B1	C1	D1	E1	F1	G1	H1
2	A2	B2	C2	D2	E2	F2	G2	H2
3	A3	B3	C3	D3	E3	F3	G3	H3
4	A4	B4	C4	D4	E4	F4	G4	H4
5	A5	B5	C5	D5	E5	F5	G5	H5
6	A6	B6	C6	D6	E6	F6	G6	H6
7	A7	B7	C7	D7	E7	F7	G7	H7
8	A8	B8	C8	D8	E8	F8	G8	H8

Figure I-1
The Checkerboard as a Worksheet

Each cell is identified by its "cell ID"—the column letter and row number, as shown in Figure I-1.

A complete cell ID also includes the layer letter for the cell. In most cases, though, you only have to use the column letter and row number.

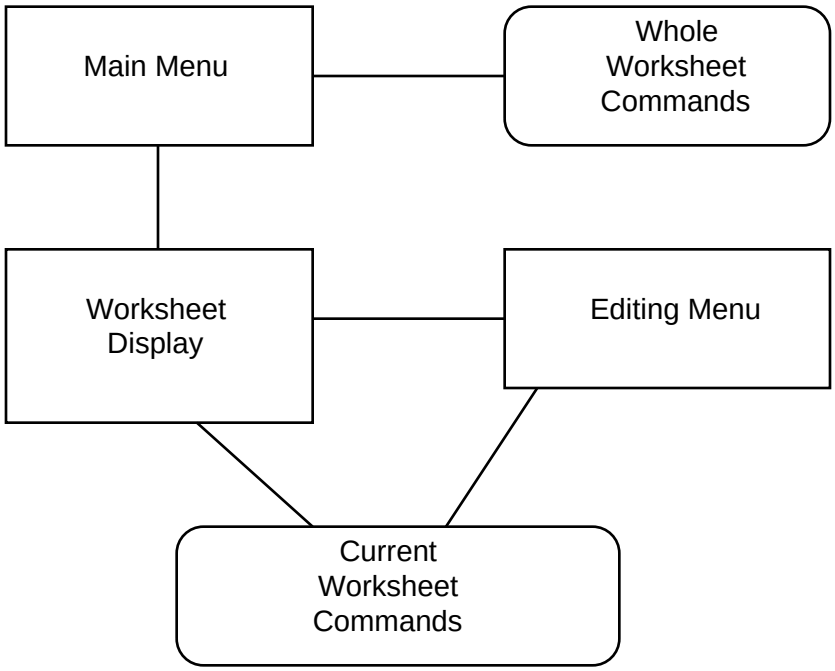
Actually, an AlphaCALC worksheet can be ***much*** larger than a checkerboard. Each layer may have up to 8191 rows and 255 columns, and a worksheet can have up to 255 layers. Since there are only 26 letters, the columns past column Z (and layers past layer Z) use two letter combinations—AA, AB, AC, and so on.

Since a worksheet can be much bigger than your terminal screen can display, AlphaCALC uses your screen as a "window" you can look through to see any part of your worksheet.

One of the things that makes AlphaCALC so powerful is the ability to refer to a value by its cell ID—its location in the worksheet. For example, if you put 10 in cell A1, you can refer to cell A1 in any other cell, and it uses the value 10. Then, when you change the value in a cell—in our example, if you change the value in A1 to 12—AlphaCALC can immediately update all the other values referring to that cell.

AlphaCALC'S DESIGN

The following diagram gives you a general idea of AlphaCALC's overall structure:



When you call up AlphaCALC, you're at the Main Menu. From the Main Menu you can go to your worksheet display or use a number of other commands which affect entire worksheets. These commands let you print, copy, or erase worksheets, see a list of worksheets, and so on.

At your worksheet display, you can enter new information or formulas, edit the data already in your worksheet, and use the function keys on your keyboard to perform editing functions, such as erasing or inserting rows and columns, and so on.

From your worksheet display, you also have access to commands which affect only the current worksheet. You can invoke these features directly from your worksheet, or by going to the AlphaCALC Editing Menu and choosing the command you want. These commands let you print your worksheet, sort sections of it, search for particular entries, and much more.

So, everything you can do in AlphaCALC, you do in one of these ways:

- Type information in a worksheet cell.
- Press a key while you're viewing your worksheet.
- Type a "slash" command from your worksheet display.
- Type a command at one of AlphaCALC's two menus.

That's a quick sketch of AlphaCALC's design. You never have to do more than press a key or two and type a word to get to the AlphaCALC feature you want. As you move through the lessons, you'll see how the combination of menus, slash commands, and special keys makes AlphaCALC very easy to use.

WHAT THIS BOOK DOES

This tutorial has two purposes:

- To teach you the basics of AlphaCALC, so you can use it right away.

- To introduce a few of AlphaCALC's advanced features. This gives you an idea of AlphaCALC's capabilities and, by exposing you to a few of the more complicated items, makes it easier for you to learn the rest of AlphaCALC's features.

After these lessons, you'll be able to use AlphaCALC in your everyday routine. But, you won't know all there is to AlphaCALC. This tutorial is really just an introduction. After you finish it, you can use the other books in the AlphaCALC Documentation Library to learn the particular AlphaCALC features you're interested in: the business math or scientific functions, the sorting capabilities, and so on.

WHO THIS BOOK IS FOR

If you've never used AlphaCALC before, this book is for you. However, it's helpful if you're familiar with the following before you start:

- The basics of your computer. If you know how to log on to your Alpha Micro system, and how to use your keyboard—for example, where the RETURN key is—you know enough to start using AlphaCALC.

- A little basic math and accounting. The sample worksheet you create and work on throughout this tutorial is a simplified income statement for a fictitious company. If you've never worked with an income statement before, pay special attention to the "About the Sample Worksheet" section at the end of this introduction.

If you're an experienced AlphaCALC user, you probably don't need to do the lessons here, though they could help you brush up if it's been a while since you used AlphaCALC. Instead, you can find the information you need in the *AlphaCALC Reference Book*.

HOW TO USE THIS BOOK

This tutorial consists of five lessons. You can do them in one session, one at a time, or in whatever sections fit your schedule. After you finish Lesson 1, you know how to start and leave AlphaCALC, and store the work you've done, so you can stop in the middle of a lesson and pick up later where you left off.

While you can stop and start your work at any time, you do need to do the lessons in order. Each lesson builds on what you learned and did previously, so you can't skip ahead.

THE STRUCTURE OF THE TUTORIAL

This book contains five lessons, each leading you through one or more AlphaCALC features. This list summarizes each lesson:

- LESSON 1 - "Getting Started." Starting and leaving AlphaCALC; AlphaCALC's menus; creating and storing a worksheet.

- LESSON 2 - "Entering Data." How to return to the worksheet you created in Lesson 1; moving around your worksheet; entering text, numbers and math formulas in worksheet cells.

- LESSON 3 - "Printing Your Worksheet." How to print your worksheet.

- LESSON 4 - "Advanced Data Entry." More ways to move around your worksheet; changing data in cells; recomputing your entire worksheet; changing how data appears; an advanced math function.

- LESSON 5 - "Rearranging Your Worksheet." Inserting, deleting and copying sections of your worksheet; searching for and replacing a particular cell entry.

After you finish the lessons, if you feel you want more experience, you can continue with Appendix A - "For More Practice," which has you enter more data and more complex formulas, and use many of the features you've already learned, plus a few new ones.

At the end of the Tutorial is a glossary which lists special AlphaCALC terms and computer terms you may not be familiar with. If you run across a word or phrase you don't know, you can look it up here.

HOW THE LESSONS ARE ORGANIZED

Each of the five lessons follows the same format. A short introduction at the beginning explains what you do in the lesson—create a worksheet, print what you've done, and so on.

Each lesson consists of a series of steps leading you through AlphaCALC's features. Each step has several parts:

- A brief introduction describing what you do in this step.
- Instructions telling you exactly what you need to type or what key you need to press to continue. Each instruction starts with a word in bold capitals, such as **TYPE** or **PRESS**.
- What the result of your action is—what appears on your screen. These descriptions begin with the words **YOU SEE** in bold capitals.
- Explanations—why you type a particular instruction, or why AlphaCALC responds like it does.
- Many steps also include illustrations, showing what your terminal screen looks like at this point.
- At the end of each step is a brief summary of what you've accomplished.
- After each step is the section "What Do I Do If?" This section lists error messages and other "unusual" conditions that could occur—and what to do if they happen. If you see anything you don't expect, turn to this section and follow the directions given.

Each step may also include other information, such as hints, remarks, and, occasionally, warnings. These optional sections contain information related to—but not actually part of—the action you're performing.

PAGE FORMAT AND CONVENTIONS

As the previous section describes, there are many types of information in this tutorial. We've designed this book so different types are as easy to tell apart as possible—so you can tell at a glance whether we are talking about a command or key, when you're to type an entry, and so on.

CONVENTION	EXPLANATION
AlphaCALC Main Menu	AlphaCALC screen names are printed in the same type face as regular text, with initial capitals.
WHAT YOU TYPE	Text you are expected to type is shown in a bold Courier typeface. If it's all capitals, type it exactly as shown. If it's in upper and lower case, It's a description of what you need to enter (for example, **Your name**).
What AlphaCALC Displays	Text AlphaCALC displays on your screen is shown in Courier typeface.
PRINT	AlphaCALC command names are shown in small, bold capitals.
KEY	Keys you need to press are shown in a keycap box.
word	Important terms you may not be familiar with are in bold italic type. You can look these words up in the glossary at the back of the book.
Text	Especially important text, stating a general concept or something you need to know throughout AlphaCALC, is enclosed in a box and printed on a gray background.
COMMAND	AMOS command names and worksheet names, such as INCOME, are shown in all capitals.
FUNCTION	AlphaCALC function names are also in all capitals.
	The halt symbol marks important information you should read carefully before continuing. This often indicates something you MUST or MUST NOT do, so read carefully.

CONVENTION	EXPLANATION
	The hint symbol marks a helpful bit of information. It often points out a short cut which may save you time or trouble.
	The remember symbol marks information you need to keep in mind while you follow a set of instructions.

For example, this instruction means to type CREATE and press the RETURN key:

CREATE RETURN

These conventions are the same as those in the rest of the AlphaCALC Documentation Library.

A NOTE ON SCREEN ILLUSTRATIONS

Your screen may not always look exactly like the sample screens shown in this tutorial. Possible reasons for this include:

- Different types of terminals sometimes include different information on the screen. For example, some terminals display the worksheet name at the top of your screen while you're viewing your worksheet, while others do not.
- You may be using a slightly different version of the AlphaCALC software. This tutorial applies to AlphaCALC versions 2.0 and later. All the screen illustrations were created using version 2.0. Later versions of AlphaCALC may have small changes to the screen displays.
- You can enter text in AlphaCALC in either upper or lower case, or any combination. Use whatever case is easiest and most attractive to you.

If there are slight differences in the screen displays, don't worry. As long as the information is the same, small format differences won't cause any problems.

SOME NOTES ON KEYS

To use many of AlphaCALC's features, you press a specific key on your keyboard. This may be a "special" key, such as the RETURN or HELP key, or it may be one of the function keys many keyboards have along the top or left of the keyboard. These function keys are labeled with the letter F and a number.

For the terminals normally used with AlphaCALC, Alpha Micro provides a plastic overlay showing what each function key does.

Because different function keys perform different features on different terminals—for example, the [MENU] key is [F3] on some terminals and [F13] on others—we always refer to keys by the *feature* name, not the key number.

If your terminal doesn't have function keys, or if it doesn't have one of the other keys we mention, such as a [HELP] key, don't worry. You can still use all of AlphaCALC's features. If you don't have a particular function or special key, you can use the "universal key sequence" for the feature. This is a sequence of two or three keystrokes, using either the [CONTROL] key or the [ESCAPE] key.

If you don't have the key for a feature you want to use, refer to the *AlphaCALC Keyboard Command Quick Reference Card* to find the alternate key for your terminal.

ABOUT THE SAMPLE WORKSHEET

The sample worksheet you create in this tutorial is a simplified income statement for a fictitious company, Wizmo, Inc. The income statement lists the various categories of income and expenses for the company, and calculates the company's gross profit margin and operating income.

Generally, income statements have one section for the current month, and another for the yearly totals. In this tutorial, you enter only the monthly figures. The yearly information, along with other sections you can add to the income statement, is in Appendix A - "For More Practice."

You enter this information in stages, starting with section titles and continuing with numeric values and math calculations. After you enter all the information, you go back and improve the worksheet's appearance by changing the way the values appear and adding spaces and lines where appropriate.

Once you're familiar with AlphaCALC, you probably won't follow this "two step" method: entering data, then going back to correct the appearance. Instead, you'll put the information in the format you want as you enter it. However, separating these procedures lets us introduce AlphaCALC's various features one at a time.

LESSON ONE
GETTING STARTED

Welcome to AlphaCALC! During this lesson, you learn how to:

- Call up AlphaCALC.
- Use AlphaCALC's **HELP** command, which lets you view information about any AlphaCALC feature, right on your terminal screen.
- Create an AlphaCALC worksheet.
- Store the worksheet so you can return to it.
- Leave AlphaCALC.

Before you start, make sure you have an AlphaCALC keyboard overlay for your terminal, showing what each of the function keys on your keyboard does in AlphaCALC. When you need to press a key, we refer to it by the feature name on the overlay, not by the F number on the function key. If you don't have a keyboard overlay, keep the *AlphaCALC Keyboard Command Quick Reference Card* handy so you can see which key on your terminal performs any feature you need to use.

1 CALLING UP AlphaCALC

In this step, you learn how to start using AlphaCALC.

Before you start these instructions, make sure you're logged on to your computer, in the account where you want to use AlphaCALC. If you don't know how to log on, or if you're not sure what account to use, see your System Operator. Once you're logged on, and at the AMOS prompt, follow the instructions below.

These instructions assume you start from the ***AMOS prompt***. However, your computer system may display a menu when you log on. If so, AlphaCALC should be one of the choices. To start using AlphaCALC, select it from the menu.

There is also a chance you'll call up AlphaCALC by using a command other than CALC, such as ACALC. If you're not sure how to start AlphaCALC, see your System Operator.

TYPE: **CALC** RETURN

YOU SEE: After a few seconds, the AlphaCALC Main Menu, as shown in Figure 1-1.

```
            #           #                   #########           #
            #           #                   #                   #
   #######  #  #######  #######  #######    #          #######  #  #######
         #  #  #     #  #     #        #    #                #  #  #
   #######  #  #     #  #     #  #######    #          #######  #  #
   #     #  #  #     #  #     #  #     #    #          #     #  #  #
   #######  #  #######  #     #  #######    #########  #######  #  #######
               #
                  Alpha Micro's Advanced Electronic Worksheet
                              Version 2.0(102)

HELP      - display help information     COPY      - make a copy of a worksheet
                                         ERASE     - erase a worksheet
CREATE    - create a new worksheet       RENAME    - rename a worksheet
EDIT      - edit an existing worksheet
DIRECTORY - display lists of worksheets  MODE      - update CALC.INI commands
                                         IMPORT    - import worksheet data
PRINT     - send worksheet to printer    EXPORT    - export worksheet data
QUEUE     - printer queue maintenance
LOAN      - loan amortization            QUIT      - leave AlphaCALC

Enter command name and press RETURN:
             Choose a command -OR- Press HELP For More Information
```

Figure 1-1
The AlphaCALC Main Menu

You are now in AlphaCALC. The Main Menu is your starting point. From the Main Menu, you can use any command listed on the lower half of your screen.

WHAT DO I DO IF?

I see other messages before the AlphaCALC Main Menu appears?

Check with your System Operator. Getting a message while you're calling up AlphaCALC means an error occurred while AlphaCALC was checking to see which options are set for your account.

Probably, this is nothing serious, and you can go ahead with the lesson after your System Operator makes a correction.

2 VIEWING HELP INFORMATION

In this step, you use the **HELP** command to see information about the **CREATE** command. Whenever you have a question about using AlphaCALC, you can view help information on the topic you want.

There are two ways to choose a command from the Main Menu or Editing Menu. The one you'll use first is typing the command name and pressing [RETURN]. The prompt at the bottom of the menu is called the ***command line***, since that's where you can enter commands. We'll show you the other way to choose commands later in the tutorial.

In case of a typing mistake, use the [RUBOUT] key to erase the mistake, then retype the command. [RUBOUT] backs the cursor up and erases what you typed.

YOU SEE: The AlphaCALC Main Menu.

TYPE: **HELP** [RETURN]

YOU SEE: The AlphaCALC Information Menu.

The Information Menu shows a list of topics, with a prompt where you pick the number you want—in this case, "Creating a New Worksheet."

TYPE: **3** [RETURN]

YOU SEE: Creating a Worksheet, Screen 1 of 4.

READ: The information on your screen.

The help information includes more details than you need to create your first worksheet. You don't need to remember them all. The important thing is to become familiar with the process of creating a worksheet.

PRESS: [NEXT SCREEN]

YOU SEE: Screen 2 of 4.

READ: Each screen of information, pressing [NEXT SCREEN] to go to the next one. If you want to back up to an earlier screen, press [PREV SCREEN].

> When you're looking at help information, you can go back and forth through the screens as many times as you want to.

PRESS: [MENU].

YOU SEE: The Information Menu.

After you view help information, you return to the Information Menu.

PRESS: [MENU]

After you use a command, you return to the menu you started from—in this case, the Main Menu.

You now know how to view AlphaCALC's help information screens. You can see information about any Main Menu command, or any Editing Menu command, because help is available from the Editing Menu, too. Later, you see how to get help from any location in AlphaCALC.

WHAT DO I DO IF?

I see the message "The item you selected is not recognized - please try again?

AlphaCALC doesn't understand what you typed on the command line. Press any character key to clear the message. Then, check for a typing error and retype the command.

The help information I see is not for the create command?

One of two things happened. Either:

- You entered a number other than 3 on the Information Menu. Or,
- You pressed [↑] or [↓] before you pressed [RETURN]. These keys let you highlight the various help selections, as part of an alternate way of choosing the topic you want. You'll learn about this method of making selections from a menu later in the tutorial.

In either case, to correct the problem, press [MENU] to return to the Information Menu, then type **3** and press [RETURN] to call up the **CREATE** help screens.

3 CREATING A WORKSHEET

Now that you've read about creating a worksheet, it's time to do it. In this step, you use the **CREATE** command to name your new worksheet. The sample worksheet is an income statement for a fictitious company.

YOU SEE: The AlphaCALC Main Menu.

TYPE: **CREATE** [RETURN]

YOU SEE: The Create a New AlphaCALC Worksheet screen, as shown in Figure 1-2. The cursor is at the prompt `Worksheet to create`.

This is an example of a "command screen." Most AlphaCALC commands display screens like this, where you enter the information for the command.

```
                    Create a new AlphaCALC worksheet

                Worksheet to create: ______



    [ Press UP ARROW or DOWN ARROW to display a worksheet directory ]

[ Press MENU to cancel the CREATE operation and return to the Main Menu ]



        Enter worksheet name -OR- press arrow keys to see directory
```

Figure 1-2
The Create a New AlphaCALC Worksheet Menu

AlphaCALC worksheet names can be up to six characters long. You can use any combination of letters and numbers for the name, but no other characters.

Generally, it's a good idea for the worksheet name to be related to its contents. That way, it's easier to remember, and easier to figure out if you happen to forget it.

TYPE: **`INCOME`** [RETURN]

YOU SEE: A new prompt, `Prototype worksheet to use: DEFWKS`.

The prototype worksheet is used as a basis for the worksheet you create. The default prototype worksheet, DEFWKS, contains only standard formatting information.

PRESS: [EXECUTE]

> On any command screen, the [EXECUTE] key tells AlphaCALC you want to perform the command.

YOU SEE: The Worksheet Header.

You assigned a name to your new worksheet. In the next step, you use the Worksheet Header to enter some general information about the worksheet.

WHAT DO I DO IF?

I see the message "The item you selected is not recognized - please try again?

AlphaCALC doesn't understand what you typed on the command line. Press any character key to clear the message. Then, check for a typing error and retype the command.

I make an error while typing the worksheet name?

If you notice a typing error before you press [RETURN], use the [RUBOUT] key to back up and retype the name.

If you don't notice the error until after you press [RETURN], press [BACK TAB] to return to the `Worksheet to create` field and retype **`INCOME`**.

I see the message "File already exists - press any key to continue"?

Someone has already created a worksheet named INCOME in this account. Press any character key to clear the message, then press [MENU] to return to the Main Menu. Talk to your System Operator, who can tell you what account you should use.

My terminal beeps while I'm entering the worksheet name?

Worksheet names can contain only letters and numbers. If you type an illegal character, AlphaCALC beeps your terminal to let you know you made a mistake.

This also happens if you type more than six characters as the worksheet name. If necessary, use [RUBOUT] to back up, then retype the name.

4 THE WORKSHEET HEADER

The AlphaCALC Worksheet Header is a form describing and defining your worksheet. You type information in each field—a description, the author's name, and so on—like you fill in the blanks on a printed form. The Worksheet Header also contains printing format settings for your worksheet, but you won't change these settings in this lesson.

Besides describing the Worksheet Header, this step explains how you can move the cursor to make changes and corrections—information which applies throughout AlphaCALC. This could come in very handy later!

YOU SEE: The Worksheet Header, with the cursor in the first field, `Description`.

You'll notice there are already entries in many of the Worksheet Header fields. Most of the entries are taken from the prototype worksheet you used. The entries in `Author` and `Operator` are your current user name.

Entries AlphaCALC fills in automatically are called ***defaults***. Many AlphaCALC commands use the most common entry in a field as the default, to save you the effort of typing it. AlphaCALC uses these default values unless you change them.

You can use upper or lower case letters in any of the Worksheet Header fields—or anywhere in AlphaCALC. Use whatever looks best and is easiest for you.

TYPE: **`INCOME STATEMENT FOR WIZMO, INC.`** [RETURN]

YOU SEE: What you type replaces the default entry. The cursor moves to the `Author` field. This field contains your current user name.

TYPE: **`Your name`** [RETURN] (You don't need to do this if it's the same as your user name.)

YOU SEE: The cursor moves to the `Operator` field. This also contains your user name.

PRESS: [CHAR DEL]

YOU SEE: The first letter of your user name is deleted.

PRESS: [CHAR DEL] to erase all of your user name.

YOU SEE: Each time you press [CHAR DEL], one character is deleted and the rest of the field moves one space to the left.

TYPE: **`Your initial, a period, and your last name`** [RETURN]

YOU SEE: The cursor moves to the `Dept` field.

PRESS: [BACK TAB]

YOU SEE: The cursor returns to the beginning of the `Operator` field.

> On any command screen, [BACK TAB] moves the cursor to the previous field, and [TAB] or [RETURN] moves it to the next field. This lets you move the cursor easily to any field you want to change.

PRESS: The [→] key.

YOU SEE: The cursor moves to the right, but doesn't erase what you typed.

MOVE: The cursor to the period after your first initial, using [→] and [←].

PRESS: [CHAR DEL]

YOU SEE: The period is deleted and your last name moves left one character.

PRESS: [CHAR INS]

YOU SEE: A space is inserted at the cursor position.

PRESS: [CHAR INS] until there are enough spaces to add your first name.

TYPE: The rest of your first name.

YOU SEE: Your name appears as you type.

PRESS: [TAB]

YOU SEE: The cursor moves to `Dept`.

Table 1-1 summarizes the ways you can change entries and make corrections. These methods work on any AlphaCALC screen, including when you're entering information in a cell.

TO DO THIS	USE THIS KEY
Back up, erasing characters	[RUBOUT]
Go forward, erasing characters	[SPACE BAR]
Back up, without erasing	[←]
Go forward, without erasing	[→]
Insert a space	[CHAR INS]
Delete a character	[CHAR DEL]

Table 1-1
Changing Entries

PRESS: [RETURN] or [TAB]

YOU SEE: The cursor goes to the `Protection` field.

You just skipped the `Dept` field. You can skip any Worksheet Header field by pressing [RETURN] or [TAB].

The `Protection` field lets you assign a password to keep other people from using your worksheet. You won't use this field now.

At this point, your screen looks like Figure 1-3.

```
 ----------------------------  Worksheet Header  ------------------------------+
!                                                                               !
!Description: Income Statement for Wizmo, Inc.                                 !
!Author: Lee Joseph             Operator: Lee Joseph             Dept:         !
!Protection: None                                                              !
!                                                                               !
!                                                                               !
+--------------------- Page Format and Print-out Options ----------------------+
!                                                                               !
!Lines per Inch - spacing...... 6    Title Range..................             !
!Page length - lines........... 66   Lines after Page Title........            !
!Top Margin - lines............ 3    Footer Range.................             !
!Bottom Margin - lines......... 4    Lines before Page Footer......            !
!Character Spacing - pitch..... 10   Print column border........... No         !
!Left Paper Offset - margin.... 10   Print row border............... No         !
!Characters per Line........... 80   Default Form................. NORMAL     !
!                                    Page Orientation............... Portrait   !
+-------------------------------  STATISTICS  -------------------------------+
!Created: 10-Jun-91 02:57 PM                                                   !
!Revised: 10-Jun-91 02:57 PM                           Total Time Open:    0:00!
!                                                                               !
!                                                                               !
+---[ Press TAB or BACK-TAB to select fields; Press EXECUTE when finished ]----+
  Worksheet protection option. Enter "N" for None, "P" for Password protection
```

Figure 1-3
The Worksheet Header Screen

So far, you created a worksheet and described it in the Worksheet Header. You also know how to move the cursor back and forth to make changes or corrections. In the next step, you view the worksheet display—where you actually enter data—and learn how to go to the AlphaCALC Editing Menu.

WHAT DO I DO IF?

My terminal beeps while I'm making an entry?

You typed too many characters in the field. The `Author` and `Operator` fields are only 20 characters long. If your name is longer than this, you'll have to abbreviate, or just use part of it. Sorry.

I accidentally go past the "Protection" field?

As long as you didn't change any entries past `Dept`, there's no problem. Just continue with the tutorial.

5 THE WORKSHEET DISPLAY AND THE MENU KEY

The worksheet display is where you do most of your work in AlphaCALC. It's where you enter information in your worksheet and tell AlphaCALC what you want it to do with the information.

In this step, you view the worksheet display and use the MENU key to go back and forth between the worksheet display and the AlphaCALC Editing Menu. MENU is one of the most important keys in AlphaCALC. It allows you to move back and forth among the worksheet display, AlphaCALC's menus, and command screens.

YOU SEE: The Worksheet Header.

PRESS: EXECUTE

YOU SEE: Your worksheet display, which looks like Figure 1-4.

```
        |    A    |    B    |    C    |    D    |    E    |    F    |    G    |    H
    1
    2
    3
    4
    5
    6
    7
    8
    9
   10
   11
   12
   13
   14
   15
   16
   17
   18
   19
Formula:
Format: GEN General
Alignment: General      F/B Color: 0/0

Cell A|A1                                                  10-Jun-91
```

Figure 1-4
The Worksheet Display

The middle of your screen shows your worksheet cells, with the column letters listed along the top and the row numbers down the left side.

There's a large cursor in cell A|A1. This cursor marks your current location in your worksheet. The cell with the cursor is called the ***current cell***.

The bottom part of your screen contains several fields describing the current cell—in this case, cell A|A1. These are the cell editing fields. The cell ID of the current cell is at the lower left of your screen.

On some terminals, your worksheet name and description appear at the top of the screen.

You learn more about your worksheet display in Lesson 2, when you start entering data.

Now, go to the AlphaCALC Editing Menu.

PRESS: MENU

YOU SEE: The AlphaCALC Editing Menu, as shown in Figure 1-5.

```
                      AlphaCALC 2.0(102)   EDITING MENU

Monday, June 10, 1991 03:02 PM           9384 bytes in use,  432808 bytes free.

Editing INCOME     Income Statement for Wizmo, Inc.

HELP      - display help information     PAGINATE  - repaginate the worksheet
FINISH    - write out current worksheet  PRINT     - send worksheet to printer
SAVE      - update current worksheet     QUEUE     - printer queue maintenance
QUIT      - quit worksheet w/o update    WRITE     - generate AlphaWRITE file

SEARCH    - locate a string              DIRECTORY - display list of worksheets
REPLACE   - replace a string             SORT      - sort information
GLOBAL    - replace all occurrences      WIDTH     - change display width

HEADER    - display header information   CUT       - copy block of cells to file
DUPLICATE - duplicate block of cells     PASTE     - copy file to worksheet
PROTECT   - block cell protection        IMPORT    - import worksheet data
FORMAT    - set format information       EXPORT    - export worksheet data
MODE      - update CALC.INI commands     LOAN      - loan amortization


Enter command name and press RETURN:

Cell A|A1                                            10-Jun-91
```

Figure 1-5
The AlphaCALC Editing Menu

> From your worksheet, pressing MENU takes you to the Editing Menu.

The Editing Menu is similar to the AlphaCALC Main Menu. It has a list of commands you can use, and a command line at the bottom where you can enter them. The Editing Menu also shows the name and description of your worksheet, and other information.

PRESS: MENU

YOU SEE: Your worksheet display.

> From the Editing Menu, pressing MENU always takes you back to your worksheet.

You now know what the worksheet display looks like, and how to go back and forth between your worksheet and the AlphaCALC Editing Menu. You'll find you use the [MENU] key more often than any key except [RETURN] and [EXECUTE].

WHAT DO I DO IF?

I see an entry in the Formula field at the bottom of my screen?

This happens if you type a character while you're at your worksheet display—AlphaCALC thinks you're starting an entry in a cell. Press [MENU] and the entry disappears.

The Editing Menu doesn't appear on my screen when I press MENU?

If you started a cell entry (see the previous explanation), [MENU] cancels the entry, but doesn't leave your worksheet display. Press [MENU] again to go to the Editing Menu.

6 STORING YOUR WORKSHEET

As you work in AlphaCALC, the information you enter is stored in your computer's temporary memory. This step shows how to store your worksheet in your computer's permanent memory so you can return to it later. The **FINISH** command from the Editing Menu does two things:

- Stores your worksheet in your computer's permanent memory.
- Returns you to the AlphaCALC Main Menu.

You can store and leave your worksheet at any time, and retrieve it when you want to add or change anything.

YOU SEE: The worksheet display.

PRESS: [MENU]

YOU SEE: The Editing Menu.

TYPE: **`FINISH`** [RETURN]

You only need to type enough of the command name to distinguish it from any other command. For example, for **FINISH**, you can type just **`FI`**, since no other command starts with FI.

YOU SEE: The AlphaCALC Main Menu.

That's all there is to storing your worksheet—use the Editing Menu **FINISH** command. You are now back at the Main Menu, and can use any Main Menu command you want.

WHAT DO I DO IF?

I see the message "The item you selected is not recognized - please try again?

AlphaCALC doesn’t understand what you typed on the command line. Press any character key to clear the message. Then, retype the command.

7 LEAVING AlphaCALC

Now, you can leave AlphaCALC and return to AMOS command level—or wherever you entered AlphaCALC from. You can only leave AlphaCALC from the Main Menu.

YOU SEE: The AlphaCALC Main Menu.

TYPE: **QUIT** [RETURN]

YOU SEE: The AMOS command prompt.

There is also a QUIT command on the Editing Menu. It does ***not*** do the same thing as Main Menu QUIT. The Editing Menu QUIT command is discussed in Lesson 5. When you want to leave AlphaCALC, you ***must*** return to the Main Menu, and use the QUIT command from there.

If your system is set up to show you a menu instead of the AMOS prompt, your menu displays on your screen. You are now out of AlphaCALC.

WHAT DO I DO IF?

I see the message "The item you selected is not recognized - please try again?

AlphaCALC doesn’t understand what you typed on the command line. Press any character key to clear the message. Then, check for a typing error and retype the command.

I see the prompt "Are you SURE you want to leave and abandon all your changes?" after I enter the quit command?

You’re still at the Editing Menu instead of the Main Menu. Press [RETURN] to return to the command line, then go back to step 6 to store your worksheet and return to the Main Menu.

LESSON TWO
ENTERING DATA

You now have a worksheet you can enter data in. The steps of this lesson show you:

- How to call up your existing worksheet.
- How to move the cursor around your worksheet.
- How to use the HELP key to see help information from your worksheet display.
- How to enter numbers in the worksheet.
- How to enter text.
- A way to speed up data entry.
- How to have AlphaCALC perform math calculations for you.
- How to use SUM, an AlphaCALC function.

1 CALLING UP YOUR WORKSHEET

The last thing you did in Lesson 1 was leave AlphaCALC. In this step, you go back into AlphaCALC and use the **EDIT** command to return to the INCOME worksheet you created earlier.

YOU SEE: The AMOS prompt.

Remember, if you see a menu instead of the AMOS prompt, or call up AlphaCALC using a different command name, follow your System Operator's instructions for starting up AlphaCALC.

TYPE: **CALC** RETURN

YOU SEE: The AlphaCALC Main Menu.

You call up an existing worksheet using the **EDIT** command. You could do this by typing EDIT (or just ED), but we'll show you another way to choose the command you want.

PRESS: [↑]

YOU SEE: The **QUIT** command line is highlighted.

PRESS: [↑] again.

YOU SEE: The **EXPORT** command line is highlighted.

You can use the [↑] and [↓] keys to highlight any command on the Main Menu. Once the command you want is highlighted, you select it by pressing [EXECUTE].

PRESS: [↑] or [↓] until the **EDIT** command is highlighted.

PRESS: [EXECUTE]

YOU SEE: The AlphaCALC Edit Existing Worksheet Menu, as shown in Figure 2-1.

```
Edit an existing AlphaCALC worksheet

Worksheet to edit: ______

[ Press UP ARROW or DOWN ARROW to display a worksheet directory ]

[ Press MENU to cancel the EDIT operation and return to the Main Menu ]

Enter worksheet name -OR- press arrow keys to see directory
```

Figure 2-1
The Edit Existing Worksheet Menu

YOU SEE: The cursor is at the prompt, `Worksheet to edit`.

All you have to do to call up your worksheet with **EDIT** is type its name.

TYPE: **`INCOME`** [EXECUTE]

YOU SEE: A message appears briefly, telling you AlphaCALC is loading your worksheet. Then, after a few seconds, your worksheet appears on your screen. When you **EDIT** a worksheet, you go straight to your worksheet display, not to the Worksheet Header.

To look at the Worksheet Header of an existing worksheet, you can use the **HEADER** command from the Editing Menu.

You called the INCOME worksheet back to your screen. In the next step, you learn how to move the cursor around your worksheet.

WHAT DO I DO IF?

I see other messages before the AlphaCALC Main Menu appears?

Check with your System Operator. Getting a message while you're calling up AlphaCALC means an error occurred while AlphaCALC was checking to see which options are set for your account.

Probably, this is nothing serious, and you can go ahead with the lesson right away or after your System Operator makes a correction.

I see the message "The item you selected is not recognized - please try again?"

AlphaCALC doesn't understand what you typed on the command line. Press any character key to clear the message, then check for a typing error and retype the command.

I type the worksheet name wrong?

If you notice a typing error before you press [RETURN], use the [RUBOUT] key to back up and retype the name.

I see the message "File not found ... press any key to continue"?

One of three things has happened:

- You typed the worksheet name wrong. If so, press a key to clear the message, then retype **`INCOME`** and press [EXECUTE].
- You're not in the same account you were when you created the worksheet. If you type INCOME correctly and get this message, press a key to clear the message, then press [MENU] to return to the Main Menu, use **QUIT** to leave AlphaCALC and log in to the proper account.
- Someone else may have erased INCOME from your account. If so, you need to start over and create the worksheet again. Sorry.

My terminal beeps while I'm entering the worksheet name?

Worksheet names can contain only letters and numbers. If you type an illegal character, AlphaCALC beeps your terminal to let you know you made a mistake.

This also happens if you type more than six characters as the worksheet name. If necessary, use [RUBOUT] to back up, then retype the name.

2 MOVING AROUND YOUR WORKSHEET

As mentioned in Lesson 1, the cursor—the lighted block in one cell—marks your current location in your worksheet. Information you enter always goes in the cell where the cursor is. In this step, you learn how to move the cursor to any cell in your worksheet.

YOU SEE: Your worksheet display with the cursor in cell A|A1.

A|A1 is a ***cell ID***. The cell ID contains the layer letter, the | character, the column letter, and the row number. So cell A|A1 is the cell in layer A, column A and row 1.

PRESS: [→]

YOU SEE: The cursor moves to cell B1, and the cell ID at the bottom left of your screen changes to A|B1.

PRESS: [↓]

YOU SEE: The cursor moves to cell A|B2.

PRESS: Any of the arrow keys, moving the cursor around your screen.

YOU SEE: The cursor moves one cell each time you press an arrow key, and the cell ID at the bottom left of your screen changes to show its current location.

When you press an arrow key, the cursor moves one cell in the direction of the arrow. You can use the arrow keys to move the cursor to any cell in the layer. However, there are faster ways.

PRESS: [GO TO]

YOU SEE: A prompt appears: `Move to Cell:`.

TYPE: **`G9`** [EXECUTE]

YOU SEE: The cursor moves to cell G9.

Because you entered only the column letter and row number, the cursor goes to cell G9 of the current layer, layer A.

> Whenever you want to refer to a cell on the current layer, you can leave the layer letter out of the cell ID.

You can move the cursor to any cell by pressing [GO TO] and typing the cell ID.

PRESS: [HOME]

YOU SEE: The cursor returns to cell A1.

[HOME] always returns the cursor to cell A1 of the current layer.

Your worksheet is much larger than your screen display. You can use any of these methods to move your cursor to cells you can't see on your screen. When you do, your screen display changes to show the cell you want. You learn other ways to move your screen display around your worksheet in Lesson 4.

You can now move the cursor to any cell you want, so you can enter information anywhere in your worksheet.

WHAT DO I DO IF?

The cursor doesn't move when I press an arrow key?

If you type a character while at your worksheet display, you enter editing mode, and you can't move around the worksheet. If there are any characters in the `Formula` field at the bottom of your screen, press [MENU] to leave editing mode. You should then be able to move around.

3 THE HELP KEY

In this step, you view help information directly from your worksheet display, using the [HELP] key instead of the **HELP** command you used in Lesson 1.

YOU SEE: Your worksheet display.

PRESS: [HELP]

YOU SEE: The AlphaCALC Information Menu. The list of topics is different than it was when you used the **HELP** command from the Main Menu. These choices all have to do with your worksheet display.

> You can press the [HELP] key anywhere in AlphaCALC to see help information.

AlphaCALC has three Information Menus—one each for the Main Menu, Editing Menu, and your worksheet display. You always see the topics relevant to where you are.

From the worksheet display, [HELP] shows you an Information Menu. From anywhere else in AlphaCALC, it goes directly to help information about your current location.

TYPE: **2** [EXECUTE]

YOU SEE: Information describing how to enter a cell formula.

> A ***cell formula*** is whatever you enter in a cell. It may be a number, text, or a calculation. Whatever it is, it's called a cell formula.

There are four screens to this help topic, but the information you need now is all on this first one. After you read it, you can either go ahead and look at the rest of the screens, or press [MENU] to return to your worksheet.

YOU SEE: Any of the help screens for Entering and Editing Cell Formulas.

PRESS: [MENU]

YOU SEE: The Information Menu.

PRESS: [MENU]

YOU SEE: Your worksheet display.

In this step, you used the [HELP] key to see help information directly from your worksheet display. The information you saw explains the process for entering data—a cell formula—in a cell.

WHAT DO I DO IF?

The help information I see is not on entering cell formulas?

One of two things happened. Either:

- You entered a number other than 2 on the Information Menu. Or,
- You press [↑] or [↓] before you pressed [RETURN]. These keys let you highlight the various help topics, like you can highlight choices on the Main Menu.

In either case, press [MENU] to return to the Information Menu, then type 2 and press [RETURN].

4 ENTERING NUMBERS

Numbers are the kind of information most commonly used in AlphaCALC worksheets. In this step, you learn how to enter them.

The procedure you learn in this step holds true no matter what kind of cell formula you enter—numbers, text, or anything else. This procedure is probably the most important part of AlphaCALC for you to know. And it's easy!

YOU SEE: Your worksheet display.

MOVE: The cursor to cell C6, using any method from step 2.

TYPE: **`1150`**

YOU SEE: A small cursor appears in the `Formula` field at the bottom of your screen. The numbers appear in this field as you type.

If you make a typing mistake, you can use any of the methods discussed in Lesson 1 to correct it.

PRESS: [EXECUTE]

YOU SEE: The number you entered, 1150, appears in cell C6.

You entered data in a cell! Whenever you enter data, you use these steps:

1. Move the cursor to the cell.
2. Type the data in the formula field.
3. Press the [EXECUTE] key. [EXECUTE] tells AlphaCALC to put what you entered in the cell.

Your screen now looks like Figure 2-2.

```
     |   A   |   B   |   C   |   D   |   E   |   F   |   G   |   H
   1
   2
   3
   4
   5
   6                       1150
   7
   8
   9
  10
  11
  12
  13
  14
  15
  16
  17
  18
  19
Formula: 1150
Format: GEN General
Alignment: General      F/B Color: 0/0

Cell A|C6                                          11-Jun-91
```

Figure 2-2
Your Worksheet after Entering a Number

PRESS: [↓] to move the cursor to cell C7.

YOU SEE: The `Formula` field is blank. This field always shows information for the current cell.

TYPE: **123**

PRESS: [EXECUTE]

YOU SEE: 123 appears in the `Formula` field as you type. When you press [EXECUTE], 123 appears in cell C7.

While you're making an entry in a cell, you're in ***editing mode***. While you're in editing mode, you can't move to another cell.

To leave editing mode without entering anything in a cell—for example, if you start typing while you're in the wrong cell—press either the [CANCEL] key or the [MENU] key.

You entered data in your worksheet! In the next steps, you see it's just as easy to enter text and math calculations as it is to enter numbers.

WHAT DO I DO IF?

The cursor goes to the Format field?

You pressed [RETURN], [TAB] or [↓] instead of [EXECUTE] after typing the cell formula. Press [EXECUTE] to enter the formula in the cell. You learn how to use the `Format` field in Lesson 4.

I enter the wrong number in a cell?

With the cursor in the cell, press [EDIT] to enter editing mode, then type the correct number and press [EXECUTE]. It's easy to change an entry in AlphaCALC!

I enter a number in the wrong cell?

If you notice the cursor is in the wrong cell after you start typing but before you press [EXECUTE], press [CANCEL] or [MENU]. This takes you out of editing mode without entering anything in the cell. Then, move the cursor to the proper cell and start over.

If you press [EXECUTE] before you notice you're in the wrong cell, follow these steps:

- Move the cursor back to the cell where you made the incorrect entry.
- Press [EDIT].
- Erase the entry using [CHAR DEL].
- Press [EXECUTE].

This erases the cell formula. Then, move the cursor to the correct cell and enter the cell formula.

5 ENTERING TEXT

It's often helpful to label the values in your worksheet so their meaning is clear. In this step, you learn how to put text in a cell, and you enter titles for your worksheet.

YOU SEE: Your worksheet display.

MOVE: The cursor to cell B1.

Wizmo, Inc. is the name of the company this is the income statement for.

TYPE: **`WIZMO, INC.`**

PRESS: [EXECUTE]

YOU SEE: As you type, your entry appears in the formula field, just as when you entered a number. When you press EXECUTE, the text appears in cell B1. If the C in INC. doesn't display, don't worry. It will when you move the cursor to another cell.

That's all there is to it. You simply type in the text, just like entering a number. AlphaCALC automatically decides if your entry is a number or text.

MOVE: The cursor to cell B2.

TYPE: **JUNE 1991**

PRESS: EXECUTE

YOU SEE: The entry appears in cell B2. Your worksheet now looks like Figure 2-3.

```
      |    A    |    B    |    C    |    D    |    E    |    F    |    G    |    H
    1           WIZMO, INC.
    2           June 1991
    3
    4
    5
    6                          1150
    7                           123
    8
    9
   10
   11
   12
   13
   14
   15
   16
   17
   18
   19
Formula: June 1991
Format: GEN General
Alignment: General      F/B Color: 0/0

Cell A|B2                                            11-Jun-91
```

Figure 2-3
Your Worksheet After Entering Text

In the next step, you enter more text and numbers, and learn a way to speed up entering data.

WHAT DO I DO IF?

See the list of posible problems at the end of step 4.

6 SPEEDING UP ENTRY

In this step, you enter row and column headings, and some more values. As you do, you learn ways to make entering data faster and easier.

YOU SEE: Your worksheet display.

MOVE: The cursor to cell C4.

TYPE: **`MONTH`**

PRESS: [EXECUTE]

YOU SEE: MONTH appears in cell C4.

MOVE: The cursor to cell A5.

ENTER: **SALES** in cell A5. Follow the steps you've learned.

From here on in this tutorial, whenever you see the instruction **ENTER**, it means:

1. Move to the cell, if you're not there already.
2. Type the cell formula.
3. Press [EXECUTE]

We use this instead of repeating each step of the entry process for every cell formula.

You've now entered the column heading and one row title. Next, you're going to go down column A, entering all the row titles.

But, rather than having to press [↓] after every cell to move to the next one, you'll use "Auto-advance mode." Auto-advance mode automatically moves the cursor to the next cell after you press [EXECUTE]. It can save you a lot of time and effort when you're entering large amounts of data in a worksheet.

PRESS: [AUTO-ADVANCE]

YOU SEE: The prompt `Auto-Advance Direction: RDI` appears near the bottom of your screen.

Auto-advance mode lets you choose the directions you want the cursor to move. Since AlphaCALC worksheet are three-dimensional, you can choose three sets of directions:

- Right or left along a row.
- Up or down a column.
- In to the next layer or out to the previous one.

You pick a direction by typing its first letter. A complete direction set is three letters.

Since you're entering row titles in column A, you want the cursor to move down after each entry.

TYPE: **DRI** [EXECUTE]

This causes the cursor to move down to the next cell after an entry. At the end of a column, it will move right, to the top of the next column. If you reach the end of a layer, it will move in to the next layer.

YOU SEE: The message `<DRI>` displays at the bottom of the screen. This message stays on your screen as long as you have auto-advance turned on.

Whenever you want to turn off auto-advance mode, just press [AUTO-ADVANCE] again.

MOVE: The cursor to cell A6.

ENTER: **PRODUCT**

YOU SEE: As you type, your entry appears in the formula field, When you press [EXECUTE], PRODUCT displays in cell A6, and the cursor moves to cell A7!

ENTER: **SERVICE**

YOU SEE: After you press [EXECUTE], the cursor moves to cell A8. You can probably see already how convenient auto-advance mode can be.

ENTER: Staying in auto-advance mode, enter the following cell formulas. Notice you leave some cells blank. Use [↓] to skip over them.

In cell:	Enter:
A8	TOTAL SALES
A10	COST OF SALES
A11	PRODUCT
A12	SERVICE
A13	TOTAL C. O. S.
A15	GROSS MARGIN
A16	PRODUCT
A17	SERVICE
A18	TOTAL G. M.

You can still use the arrow keys to move the cursor while you're in auto-advance mode.

Some of these entries extend into column B. It's OK for text to overlap into the next cell as long as the next cell is blank, and the text doesn't go past the last column in your worksheet.

After you make the last entry, your screen looks like Figure 2-4.

```
      |    A    |    B    |    C    |    D    |    E    |    F    |    G    |    H
   1          WIZMO, INC.
   2          June 1991
   3
   4                      MONTH
   5   SALES
   6  Product                1150
   7  Service                 123
   8 Total Sales
   9
  10 COST OF SALES
  11  Product
  12  Service
  13 Total C.O.S.
  14
  15 GROSS MARGIN
  16  Product
  17  Service
  18 Total G.M.
  19
Formula:
Format: GEN General
Alignment: General      F/B Color: 0/0

Cell A|A19                                  <DRI>    11-Jun-91
```

Figure 2-4
Your Worksheet with Row Titles

That's the end of this step. You learned a way to speed up entering information in your worksheet, and practiced entering text. In the next step, you enter the data for these categories, and have AlphaCALC figure out the totals for you.

WHAT DO I DO IF?

See the list of posible problems at the end of step 4.

7 MATH CALCULATIONS

So far, you haven't used AlphaCALC's mathematical abilities. In this step, you have AlphaCALC do some of your work for you—adding totals and performing other calculations.

YOU SEE: Your worksheet display.

MOVE: The cursor to cell C8.

TYPE: **C6+C7**

PRESS: EXECUTE

YOU SEE: The total of C6 and C7, 1273, appears in cell C8!

That's how easy it is to add in AlphaCALC—just tell AlphaCALC what values you want to add. The values can be anything with a numeric value: numbers, cell ID's of cells containing numbers, or more complicated things like AlphaCALC functions.

There's a reason for entering the cell IDs of the numbers (C6+C7) instead of the numbers (1150+123). Using the cell IDs makes it easier to update your worksheet if one of the numbers changes. You see this feature in Lesson 4.

MOVE: The cursor back to cell C8.

YOU SEE: The value field shows 1273, while the formula field displays =C6+C7. 1273 is the cell value, and =C6+C7 (we explain the = sign below) is the cell formula:

> The ***cell formula*** is whatever you type in the cell.
>
> The ***cell value*** is the value of the cell formula.

When the cell formula is a number or text, the cell value and formula are the same. We call these kinds of formulas ***constants***, since what is displayed in the cell is the same as what you typed. When you enter a constant, nothing displays in the `Value` field, because it would be just the same as your entry.

Any cell formula where AlphaCALC has to perform a calculation or operation of some kind, such as C6+C7, is called an ***expression***. When you enter an expression, the `Value` field shows its calculated value.

While this may be confusing right now, the difference will become clear as you use AlphaCALC. Some AlphaCALC features work on cell formulas, while others work on values, and some give you a choice of constants or expressions, so it's important to understand these terms.

An = at the beginning of your cell formula marks it as an expression. AlphaCALC adds an equal sign to any expression you enter (unless you include it yourself). = is the ***type identifier*** for an expression. Under some circumstances, AlphaCALC also adds a type identifier to text entries. We talk about text identifiers in Lesson 5.

ENTER: Now, enter the rest of the data and expressions in the proper cells in column C. You can use auto-advance mode or turn it off, whichever you're more comfortable with.

In cell:	Enter:
C11	655
C12	50
C13	C11+C12
C16	C6-C11
C17	C7-C12
C18	C16+C17

To perform arithmetic functions, AlphaCALC uses the same symbols (called operators) as most calculators: + for addition, - for subtraction, * for multiplication, and / for division.

YOU SEE: If you use a lower case C in your expressions, you'll notice that AlphaCALC changes it to upper case. AlphaCALC always displays expressions in all capitals. Your worksheet now looks like Figure 2-5.

```
      |    A    |    B    |    C    |    D    |    E    |    F    |    G    |    H
    1           WIZMO, INC.
    2           June 1991
    3
    4                       MONTH
    5    SALES
    6   Product                1150
    7   Service                 123
    8  Total Sales             1273
    9
   10  COST OF SALES
   11   Product                 655
   12   Service                  50
   13  Total C.O.S.             705
   14
   15  GROSS MARGIN
   16   Product                 495
   17   Service                  73
   18  Total G.M.               568
   19
Formula:
Format: GEN General
Alignment: General      F/B Color: 0/0

Cell A|C19                                  <DRI>    11-Jun-91
```

Figure 2-5
Your Worksheet With Monthly Gross Margin

Now, you know how to add values in AlphaCALC. Later in this lesson, you use AlphaCALC's SUM function to make addition even easier.

WHAT DO I DO IF?

In addition to the symptoms listed here, refer to the list of possible problems at the end of step 4.

My entry displays in the cell instead of its value?

If you make a typing error—for example, if you enter C6[C7 instead of C6+C7—so your entry is not an expression AlphaCALC understands, AlphaCALC assumes it is a text entry. Check your entry, move the cursor back to the cell if necessary, and type the expression correctly.

Cell C8 doesn't equal 1273?

Check the following things:

- Make sure the formula for C8 is correct—C6+C7.
- Make sure C6 and C7 have the correct numbers—1150 and 123. If the numbers are wrong, or if one of them is in the wrong cell, the total in C8 will be wrong.

Correct whichever cell formula is incorrect using the procedure earlier in this section.

8 MORE PRACTICE

In this step, you enter the monthly expense information for the income statement. There is already information down to the bottom of your screen, so you have to shift your screen display to show a different part of your worksheet.

Actually, you don't have to do anything special. When you try to move the cursor past the edge of your screen, AlphaCALC scrolls your worksheet automatically to keep the current cell on your display. Enter the cell formulas below, and notice how your screen display changes.

YOU SEE: Your worksheet display.

In cell:	Enter:
A20	EXPENSES
A21	SALES
A22	MARKETING
A23	ADMINISTRATIVE
A24	TOTAL EXPENSES
A25	OPERATING INCOME
C21	255
C22	140
C23	170

YOU SEE: As the cursor moves to each new row, your screen display scrolls down the worksheet.

You've now entered all the row titles for your worksheet, and most of the data. In the next step, you learn how to use an AlphaCALC function.

WHAT DO I DO IF?

See the list of possible problems at the end of step 4.

9 THE SUM FUNCTION

So far, all the expressions you entered contained only two values. But, Cell C24 holds the total of three numbers—cells C21, C22 and C23. You could have AlphaCALC figure this by using the expression C21+C22+C23, but there's an easier way. In cell C24, you use AlphaCALC's SUM function.

YOU SEE: Your worksheet display.

MOVE: The cursor to cell C24.

ENTER: SUM(C21:C23)

YOU SEE: The total of the three values, 565, appears in cell C24.

The SUM function tells AlphaCALC to add the values of the cells in the parentheses. These values are SUM's ***argument***, and are always contained in parentheses right after the word "SUM."

The argument you used, C21:C23, is a ***cell range***. It means "All the cells between and including these two." When you want to add several numbers, this is a lot faster than typing all the cell ID's in an expression.

AlphaCALC has over eighty functions, allowing you to perform all sorts of mathematical operations.

Now, finish column C.

ENTER: In cell C25, enter C18-C24.

YOU SEE: 3 appears in the cell. Your worksheet now looks like Figure 2-6.

```
        |    A    |    B    |    C    |    D    |    E    |    F    |    G    |    H
    8   Total Sales            1273
    9
   10   COST OF SALES
   11    Product                655
   12    Service                 50
   13   Total C.O.S.            705
   14
   15   GROSS MARGIN
   16    Product                495
   17    Service                 73
   18   Total G.M.              568
   19
   20   EXPENSES
   21     Sales                 255
   22   Marketing               140
   23   Administrative          170
   24   Total Expenses          565
   25   OPERATING INCOME          3
   26
Formula:
Format: GEN General
Alignment: General      F/B Color: 0/0

Cell A|C26                                  <DRI>     11-Jun-91
```

Figure 2-6
Your Worksheet With Operating Income

This is the operating income for the month. For this simplified income statement, we won't worry about income taxes—wouldn't that be nice!

Now that all the monthly figures are in, you need to store what you've done and return to the AlphaCALC Main Menu.

PRESS: [MENU]

YOU SEE: The AlphaCALC Editing Menu.

As on the Main Menu, you can select any Editing Menu command either by typing it and pressing [RETURN], or by using [↑] and [↓] to highlight it, then pressing [EXECUTE].

SELECT: **FINISH**

YOU SEE: After a few seconds, the Main Menu appears.

As you learned in Lesson 1, the **FINISH** command stores your worksheet in your computer's permanent memory.

In this lesson, you learned the most common, most important procedure in AlphaCALC—how to put information in your worksheet and have AlphaCALC perform calculations with it. All of AlphaCALC's advanced business math, trigonometry and other capabilities are just variations on this procedure.

WHAT DO I DO IF?

In addition to the symptoms listed here, refer to the list of possible problems at the end of step 4.

My entry displays in the cell instead of its value?

If you make a typing error—for example, if you enter C6[C7 instead of C6+C7—so your entry is not an expression AlphaCALC understands, AlphaCALC assumes it is a text entry. Check your entry, move the cursor back to the cell if necessary, and type the expression correctly.

A cell value doesn't equal the correct number?

Check the following things:

- Make sure the cell formula is correct.
- Make sure the cells the formula refers to have the correct values. If the numbers are wrong, or if one of them is in the wrong cell, the value of any cell which refers to them will be incorrect.

Correct whichever cell formula is incorrect by moving to the cell and typing the proper entry.

LESSON THREE
PRINTING YOUR WORKSHEET

Having your data in a worksheet is all well and good, but you can't often take a terminal with you to a meeting to present your findings. In this lesson, you learn how to print your worksheet.

1 THE MAIN MENU PRINT COMMAND

The **PRINT** command is available from both the Main Menu and the Editing Menu. In this lesson, you print your worksheet starting from the Main Menu.

YOU SEE: The AlphaCALC Main Menu.

SELECT: **PRINT**, using either of the methods you've learned.

YOU SEE: The Print an AlphaCALC Worksheet screen.

Like any other command, you choose **PRINT** by typing it or highlighting it and pressing [EXECUTE]. In the next steps, you choose among the various printing options AlphaCALC offers.

WHAT DO I DO IF?

I see the message "The item you selected is not recognized - please try again"?

AlphaCALC doesn't understand what you typed on the command line. Press any character key to clear the message, then check your spelling and retype the **PRINT** command.

I see the message "No printers defined"?

AlphaCALC doesn't have any printers set up to use from your account. Check with your System Operator.

2 CHOOSING A WORKSHEET AND OPTIONS

In this step, you tell AlphaCALC what worksheet you want to print, and what printing options you want to use. You also learn about a convenience feature available throughout AlphaCALC.

YOU SEE: The Print an AlphaCALC Worksheet screen. The cursor is at the prompt, `Worksheet to print`.

PRESS: [↑]

YOU SEE: A list of the worksheets in your account.

You can view this list of worksheets, called a "directory," by pressing [↑] or [↓] any time you need to enter a worksheet name—such as during the **CREATE** or **EDIT** command.

The worksheets are listed in alphabetical order. If there are more than eleven worksheets in this account, you can view the rest of the directory by pressing [NEXT SCREEN] to scroll the list.

The cursor is on the first worksheet in the list. You can select the worksheet you want by moving the cursor to it using [↑] and [↓] and pressing [EXECUTE].

SELECT: The INCOME worksheet using [↑] and [↓]. (If INCOME is not shown in the list, use [NEXT SCREEN] until it is.)

PRESS: [EXECUTE]

YOU SEE: The name INCOME is entered in the `Worksheet to print` field and the cursor returns there.

PRESS: [EXECUTE]

YOU SEE: A message telling you AlphaCALC is loading and paginating the worksheet, then the Print Worksheet screen, as shown in Figure 3-1.

```
 -------------------  Print Worksheet  -------------------+
!                                                          !
!               Output Format: Standard                    !
!                                                          !
! Print by Cell Range or Page: Page                        !
!                                                          !
!                                                          !
!               Print Page(s):   1 through 1               !
!                                                          !
!      [ Press EXECUTE to print -OR- MENU to cancel ]      !
+----------------------------------------------------------+



S)tandard (cell values), F)ormulas Only or E)xtended (values and formulas)
```

Figure 3-1
The Print Worksheet Screen

The cursor is at the `Output Format` field. This lets you choose whether to print cell values, cell formulas, or both. For this lesson, print both.

TYPE: **E**

The other fields on this screen let you choose whether to divide the worksheet into pages, and/or print only part of your worksheet. Since the INCOME worksheet is very small, you don't need to change these fields.

PRESS: [EXECUTE]

YOU SEE: The AlphaCALC Printer Selection screen.

In this step, you learned a new way to choose the worksheet you want—one you can use during many different AlphaCALC commands. You also selected printing options for the INCOME worksheet.

WHAT DO I DO IF?

INCOME isn't included in the list of worksheets?

First, use [NEXT SCREEN] to make sure you've looked at all the worksheets in this account. If INCOME isn't listed, either:

- You’re in the wrong account. If you left AlphaCALC between lessons 2 and 3, either you or someone else may have logged this terminal to a different account. Press [MENU] to return to the Main Menu, use the **QUIT** command to leave AlphaCALC, and check to see if you’re in the right account. If not, LOG to the correct account and start the lesson over.

- If INCOME isn’t in your directory, and you are in the right account, someone else may have erased your worksheet. Sorry, but you’ll have to start over and **CREATE** the worksheet again.

I see the message "The item you selected is not recognized - please try again"?

You typed an answer the current field doesn’t accept. Press any key to clear the message, check to make sure you’re at the correct place in the step, and type the correct entry.

My terminal beeps while I’m making an entry?

Either you’ve typed an illegal character (such as a letter in a field that accepts only numbers), or you’ve typed too many characters. AlphaCALC beeps to let you know you’ve made an error. Check the field you’re in and correct your entry.

3 SELECTING A PRINTER

The AlphaCALC Printer Selection Screen lets you choose the printer you want and other printing options. In this step, you’ll choose a printer and print your worksheet.

YOU SEE: The Printer Selection Screen, as shown in Figure 3-2. The printers available to you are listed in the middle of your screen. The cursor is at the prompt `Please Select Printer`.

The list of printers in Figure 3-2 is just an example. The list on your screen will depend on the printers available on your system.

```
 ---------------------------------------------------------------------------+
!                          Please Select Printer: 1_                         !
!                                                                            !
!   1) Laser Printer                        2) The Fast Draft Printer        !
!                                                                            !
!                                                                            !
!                                                                            !
!                                                                            !
!                                                                            !
!                                                                            !
!                                                                            !
!   Send output to the printer? Yes                       Forms Type: NORMAL !
!  Erase output after printing? Yes                 Number of copies: 1      !
!                                                                            !
!                [ Press EXECUTE when finished; MENU to abort ]              !
+---------------------------------------------------------------------------+

            Please select the desired printer number from the list.
```

Figure 3-2
The Printer Selection Screen

SELECT: The printer you want by typing its number. If there is only one printer shown, or you want printer number 1, you don't need to do anything.

The fields at the bottom of the screen let you choose other printing options, such as printing the worksheet to a disk file to send to the printer later, printing multiple copies, and so on. You don't need to change any of those fields now.

PRESS: EXECUTE

YOU SEE: You return to the Main Menu.

You can now go get your printed worksheet, unless the printer you chose was busy. If something else is printing, your worksheet is put at the end of the ***printer queue***—the files waiting to print—and will print when the printer is available.

When you get your printed worksheet, you see it has two sections:

- The first page shows the cell values, and looks almost exactly like the worksheet display on your screen.
- The second page shows the cell formulas of your worksheet, listed by row, down the left side of the page.

You printed your worksheet! You can also print from the AlphaCALC Editing Menu. When you use **PRINT** from the Editing Menu, you do not enter a worksheet name—it automatically prints the worksheet you are editing. Otherwise, it works the same way as Main Menu **PRINT**.

WHAT DO I DO IF?

I see the message "The item you selected is not recognized - please try again"?

At the `Please Select Printer` prompt, you entered a number that isn't on the list of printers. Press a key to clear the message, then type a printer number from the list and press [EXECUTE].

My terminal beeps while I'm making an entry?

Either you've typed an illegal character (such as a letter in a field that accepts only numbers, or you've typed too many characters. AlphaCALC beeps to let you know you've made an error. Check the field you're in and correct your entry.

LESSON FOUR
ADVANCED DATA ENTRY

In this lesson, you learn a lot more about entering data in your worksheet. You learn:

- How to change a cell formula you've already entered.
- How to set the format of cells so the information displays the way you want.
- How to use IF, one of AlphaCALC's advanced logical functions.
- How to store your worksheet in your computer's permanent memory without returning to the Main Menu.

1 MOVING YOUR WORKSHEET DISPLAY

In Lesson 2, you learned how to move the cursor around your worksheet. In this step, you learn how to shift your screen display to other parts of your worksheet.

In this step, we assume you see nineteen rows and eight columns of your worksheet at one time. If your screen displays more or less of your worksheet, you won't see exactly the rows and columns listed below. However, these methods for moving around your worksheet still work.

YOU SEE: The AlphaCALC Main Menu.

SELECT: **EDIT**

YOU SEE: The Edit Existing Worksheet Menu.

SELECT: The INCOME worksheet.

YOU SEE: Your worksheet display, with the data you entered in Lesson 2.

You retrieved your income statement worksheet. But, you can't see all the data, because some of it goes past the bottom of your screen.

If you were in auto-advance mode when you **FINISH**ed from your worksheet in Lesson 2, you still are. AlphaCALC stores this option just like it stores the other information in your worksheet.

TURN OFF: Auto-advance if it is on (press the [AUTO ADVANCE] key to turn it off).

PRESS: [NEXT SCREEN]

YOU SEE: Your screen shows rows 20-38 instead of rows 1-19.

PRESS: [NEXT SCREEN]

YOU SEE: Rows 39-57 of your worksheet. These rows are empty.

[NEXT SCREEN] shifts your display "down" one screen. Each time you press it, you see the next 19 rows of your worksheet. The cursor stays in the same position on your screen. The current cell is now A39.

MOVE: The cursor to any cell in the middle of your screen.

PRESS: [PREV SCREEN]

YOU SEE: Rows 20-38 of your worksheet display.

[PREV SCREEN] is the opposite of [NEXT SCREEN]. It moves your worksheet display "up" one screen. Notice the cursor stays in the same relative position in the middle of your screen.

You can also move your screen display left and right.

PRESS: [SCREEN RIGHT]

Your screen still shows rows 20-38, but you now see columns I-P. The cursor stays in the same row, but moves to the far right of your screen, column P.

PRESS: [SCREEN LEFT]

YOU SEE: Cells A20 through H38. The cursor moves to the left side of your screen, column A.

[SCREEN RIGHT] always moves the display one screen to the right, and the cursor to the right edge of the screen. [SCREEN LEFT] does exactly the opposite.

In AlphaCALC, you can change the width of columns, and—on some kinds of terminals—the number of characters shown on your screen. So, [SCREEN RIGHT] and [SCREEN LEFT] don't always move eight columns—it may be only four if your columns are very wide, for example—but it is always exactly one screen.

PRESS: [PREV SCREEN]

YOU SEE: Cells A1 through H19. You're back where you started.

You now know how to move your screen display to other parts of your worksheet. There are other ways, too, some of which you use later in this lesson.

WHAT DO I DO IF?

I see the message "The item you selected is not recognized - please try again"?

AlphaCALC doesn't understand what you typed on the command line. Press any character key to clear the message, then check for typing errors and retype the **EDIT** command.

I type the worksheet name wrong?

If you notice a typing error before you press [EXECUTE], use the [RUBOUT] key to back up and retype the name.

I see the message "File not found - press any key to continue"?

One of three things has happened:

- You typed the worksheet name wrong. If so, press a key to clear the message, then retype INCOME and press [EXECUTE].
- You're not in the same account you were when you created the worksheet. If you type INCOME correctly and get this message, press a key to clear the message, then press [MENU] to return to the Main Menu, leave AlphaCALC and log in to the proper account.
- If INCOME isn't in your directory, and you are in the right account, someone may have erased it. If so, you'll have to re-create the worksheet and start over again. Sorry.

My terminal beeps while I'm entering the worksheet name?

Worksheet names can contain only letters and numbers, and can't have more than six characters. If you type an illegal character, or more than six characters, AlphaCALC beeps your terminal to let you know you've made a mistake.

If necessary, use [RUBOUT] to back up, then retype the name.

I accidentally start a cell formula entry?

If you type a character by mistake, and it appears in the `Formula` field, press [MENU] or [CANCEL] to cancel the entry and leave editing mode.

My screen doesn't show the right part of my worksheet?

If you accidentally move your screen display to another part of your worksheet, don't worry. Just use any of the methods you know to move the display back where you want.

2 CHANGING A CELL FORMULA

In Lesson 2, you entered data in empty cells, but never changed a cell once you had a formula in it. In this step, you learn how to change existing cell formulas. The procedure is very similar to entering data in an empty cell.

YOU SEE: Your worksheet display, cells A1 through H19.

In looking at the figures again, you decide the 655 you entered should have been 566. You need to change the entry.

MOVE: The cursor to cell C11.

TYPE: **566**

YOU SEE: As soon as you type the first number, the cursor appears in the formula field. As you type, the 566 replaces 655.

PRESS: [EXECUTE]

YOU SEE: 566 appears in the value field and in cell C11.

That's all there is to changing a number—retype it. Next, you change the month in cell B2.

MOVE: The cursor to cell B2.

PRESS: [EDIT]

YOU SEE: The cursor appears in the formula field, on the J in June.

You can use the [EDIT] key to enter editing mode without typing a character. Or, as you did in cell C11, you can just start typing your new entry.

PRESS: [→]

YOU SEE: The cursor moves one character to the right.

While you're in editing mode, [←] and [→] move the cursor in the formula field. This lets you change just the part of an entry you want.

MOVE: The cursor to the first 1 in 1991.

PRESS: [CHAR INSERT] three times.

YOU SEE: The 1991 moves three spaces to the right.

PRESS: [CHAR DELETE] three times.

YOU SEE: 1991 moves back to its original position.

You can use all of the ways of correcting typing errors, listed in Table 1-1, in editing mode.

TYPE: Change the `JUNE` to `MAY`.

YOU SEE: The characters appear in the formula field.

PRESS: [EXECUTE]

YOU SEE: The new entry appears in cell B2.

Now, you know how to change data in a cell. But, other values in your worksheet may depend on the value you change—in fact, other cells depend on the number you changed in cell C11. In the next step, you see how easy it is to update these other cells.

WHAT DO I DO IF?

I start editing the wrong cell formula?

If you start to change the wrong cell formula, and notice before you press [EXECUTE], press [MENU] or [CANCEL]. This takes you out of editing mode, and the cell formula does not change.

If you change the wrong cell formula, just follow the procedure in step 2 to change it back to the correct entry. Consider it extra practice.

3 RECOMPUTING YOUR WORKSHEET

One of AlphaCALC's most important features is its ability to update your worksheet quickly to reflect changing information. In this step, you update the rest of your worksheet to reflect the change you made in the last step.

YOU SEE: Your worksheet display.

PRESS: [RECOMPUTE]

YOU SEE: The amounts in cells C13, C16, C18 and C25, all of which depend directly or indirectly on the value of cell C11, change to agree with its new value! Move your screen down your worksheet to check this. Rows 8-26 of your worksheet now look like Figure 4-1.

```
        |    A    |    B    |    C    |    D    |    E    |    F    |    G    |    H
    8   Total Sales            1273
    9
   10   COST OF SALES
   11    Product                566
   12    Service                 50
   13   Total C.O.S.            616
   14
   15   GROSS MARGIN
   16    Product                584
   17    Service                 73
   18   Total G.M.              657
   19
   20   EXPENSES
   21     Sales                 255
   22   Marketing               140
   23   Administrative          170
   24   Total Expenses          565
   25   OPERATING INCOME         92
   26
Formula:
Format: GEN General
Alignment: General      F/B Color: 0/0

Cell A|A26                                          11-Jun-91
```

Figure 4-1
Your Worksheet After RECOMPUTE

The RECOMPUTE key recalculates all of the formulas in your worksheet to reflect the current values of all the cells. In a large worksheet, where dozens of values may depend on one number, this can save you hours of time.

This is why you use cell IDs in math formulas instead of entering the numbers in each one. Because you enter each number only once, and refer to it by its cell ID in each formula, you have to change only one cell when a value changes. Then, RECOMPUTE quickly updates the rest of your worksheet.

RECOMPUTE is one of the most powerful and useful features in AlphaCALC. It makes AlphaCALC perfect for "what if" forecasts, since it allows you to see in a matter of seconds how results change with different data.

4 ENTERING A PERCENTAGE

So far, all of the values you entered are just numbers. They could represent dollars, loaves of bread, or runs scored by a baseball team. However, AlphaCALC offers several cell formats, allowing you to specify the kind of value a number represents. You can't specify loaves of bread or runs scored, but you can choose dollar amounts, percentages, scientific notation and many others.

In this step, you calculate some percentages based on the information you entered.

YOU SEE: Your worksheet display.

MOVE: The cursor to cell D4.

ENTER: **%** in cell D4.

Even though % is not a letter, you can enter it as text. You can enter any character as text in a cell, including numerals. However, for numbers and certain other entries, you have to include a quotation mark (") as the first character, so AlphaCALC knows this is text, not a number. See the *AlphaCALC Reference Book*.

You now have a heading for the percentage column.

MOVE: The cursor to cell D6.

TYPE: **`C6/C8`**

PRESS: [RETURN]

YOU SEE: The cursor moves to the `Format` field.

> When you're in editing mode, [RETURN], [TAB] or [↓] moves the cursor to the next field; [EXECUTE] takes you out of editing mode and places the value in the cell.

TYPE: **`%1`**. This stands for percentage, rounded to one decimal place.

PRESS: [EXECUTE]

YOU SEE: 90.3% appears in cell D6. Product sales are slightly over 90% of total sales.

The `Format` field lets you select how you want the value in the cell to display. There are about twenty predefined numeric formats, allowing you to choose currency, scientific notation, percentages, and so on. In addition, many of the formats cause values to round to a specific number of decimal places, like %1.

The other fields below the formula field set other aspects of a cell's appearance. You'll use some of these fields later in this tutorial.

The `Value` field shows `.9033778476`. This is the exact—almost—value of 1150 divided by 1273. The value field always shows the plain value of the expression, regardless of the cell's format.

MOVE: The AlphaCALC cursor to cell D7.

TYPE: **`C7/C8`**

PRESS: [TAB]

TYPE: **%1**

PRESS: [EXECUTE]

YOU SEE: 9.7% displays in cell D7. Your worksheet now looks like Figure 4-2.

```
     |    A    |    B    |    C    |    D    |    E    |    F    |    G    |    H
  1          WIZMO, INC.
  2          May 1991
  3
  4                      MONTH      %
  5   SALES
  6   Product               1150    90.3%
  7   Service                12      9.7%
  8  Total Sales            1273
  9
 10  COST OF SALES
 11   Product                566
 12   Service                 50
 13  Total C.O.S.            616
 14
 15  GROSS MARGIN
 16   Product                584
 17   Service                 73
 18  Total G.M.              657
 19
Formula: =C7/C8
Format: %1  0.0%                      Value: 9.6622152396E-2
Alignment: General      F/B Color: 0/0

Cell A|D7                                          11-Jun-91
```

Figure 4-2
Your Worksheet With Percentages

You could continue this way, entering each formula in this column and setting the format for each cell. However, in the next step you learn an easier way to set the format for a large group of cells.

WHAT DO I DO IF?

I make an entry in the wrong cell?

If you notice the cursor is in the wrong cell before you press [EXECUTE] after the cell formula, press [MENU] or [CANCEL]. This takes you out of editing mode without entering anything in the cell. Then, move to the proper cell and start over.

If you complete an entry—that is, press [EXECUTE] to enter the cell formula—in the wrong cell, follow the procedure in step 2 to change the entry to the correct one. If the cell should stay blank, follow these steps to erase the cell formula:

1. Move the cursor back to the cell where you made the incorrect entry.

2. Press [EDIT] to enter editing mode.

3. Erase the entry using [CHAR DEL].

4. Press [EXECUTE].

This erases the cell formula. Then, move to the correct cell and enter the cell formula.

I see the message "Expression syntax is invalid" after I press EXECUTE?

AlphaCALC doesn’t understand what you typed in the cell. Retype the cell formula, making sure it’s correct.

My expression entry displays as text in the cell?

When you make an entry in a cell, AlphaCALC analyzes it to see if it is a valid expression. If it is, AlphaCALC evaluates it and puts the value in the cell. If it’s not an expression, or a number, AlphaCALC assumes it’s text and displays the text in the cell.

So, if you make a typing error when entering an expression, AlphaCALC may not be able to interpret it correctly, and so displays it as text. Check your entry and retype it, making sure it’s correct.

The cursor doesn’t go to the "Format" field?

You probably pressed [EXECUTE] after the cell formula, instead of [RETURN] or [TAB]. Start the entry over, being sure to press [RETURN] or [TAB] after the formula.

5 SETTING CELL FORMATS

In this step, you use the Editing Menu **FORMAT** command to set the format of a number of cells at once. First, enter the following cell formulas. Don’t change the format—just press [EXECUTE] after typing the formula.

You can turn on auto-advance mode to make these entries easier.

ENTER:

In cell:	Enter:
D8	D6+D7
D11	C11/C13
D12	C12/C13
D13	D11+D12
D16	C16/C6
D17	C17/C7
D18	C18/C8
D21	C21/C8
D22	C22/C8
D23	C23/C8
D24	SUM(D21:D23)
D25	C25/C8

Don't worry about the symbol # preceding the values. This symbol indicates a number is too long to fit in its cell. Since the values aren't rounded off yet, many of them have too many decimal places to fit.

YOU SEE: As you enter each formula, the value appears in the cell. When you finish, your worksheet looks like Figure 4-3.

```
      |    A    |    B    |    C    |    D    |    E    |    F    |    G    |    H
    7  Service                 123     9.7%
    8  Total Sales            1273        1
    9
   10  COST OF SALES
   11   Product                566#.91883116
   12   Service                 50#8.1168831
   13  Total C.O.S.            616        1
   14
   15  GROSS MARGIN
   16   Product                584#.50782608
   17   Service                 73#.59349593
   18  Total G.M.              657#.51610369
   19
   20  EXPENSES
   21    Sales                 255#.20031421
   22  Marketing               140#.10997643
   23  Administrative          170#.13354281
   24  Total Expenses          565#.44383346
   25  OPERATING INCOME         92#7.2270227
Formula: ==C25/C8
Format: GEN General                     Value: 7.227.227808E-2
Alignment: General      F/B Color: 0/0

Cell A|D25                                  <DRI>    11-Jun-91
```

Figure 4-3
Your Worksheet With Column D

To change the format of these cells, you need to go to the Editing Menu.

PRESS: MENU

YOU SEE: The AlphaCALC Editing Menu.

SELECT: The **FORMAT** command.

YOU SEE: The Cell Format Information screen. The cursor is in the `Cell range` field.

This screen lets you set the aspects of cell appearance for a group of cells instead of one cell at a time. In this step, you change only the cell format.

The first step is to enter the cell range you want to affect. In this case, it's cells D8 through D25, where the percentages are.

TYPE: **`D8:D25`**

YOU SEE: Your screen now looks like this:

```
                      AlphaCALC 2.0(102)   EDITING MENU

Tuesday, June 11, 1991 04:46 PM         11606 bytes in use,  430586 bytes free.

Editing INCOME     Income Statement for Wizmo, Inc.

HELP  +-------------------  Cell Format Information  --------------------+eet
FINISH!                                                                   !nter
SAVE  !   Cell Range: D8:D25____________________                          !ance
QUIT  !       Format: GEN General                                         !ile
      !    Alignment: General                                             !
SEARCH!   Clear ACUM: No                                                  !heets
REPLAC!    F/B Color: 1/0                                                 !
GLOBAL! Column Width:  9                                                   !
      !                                                                   !
HEADER!                                                                   !
DUPLIC! [ Press EXECUTE to perform format function -OR- MENU to cancel ] !o file
PROTEC+-------------------------------------------------------------------+t
FORMAT    - set format information       EXPORT    - export worksheet data
MODE      - update CALC.INI commands     LOAN      - loan amortization


Enter command name and press RETURN:
                     Enter the range for FORMAT to act upon
```

Figure 4-4
The Cell Format Information Screen

MOVE: The cursor to the `Format` field.

YOU SEE: The `Column width` field disappears. You can't set the column width for a cell range.

TYPE: **`%1`**

PRESS: EXECUTE

As with all other AlphaCALC commands, you must press EXECUTE to perform the command. In this case, pressing EXECUTE makes your changes take effect.

YOU SEE: The cursor returns to the `Cell range` field.

PRESS: MENU

YOU SEE: The Editing Menu

From an Editing Menu command, pressing MENU returns you to the Editing Menu.

PRESS: MENU

YOU SEE: Your worksheet display. All the values in column D are now in percent format and rounded to one decimal place, as Figure 4-5 shows.

```
      |    A    |    B    |    C    |    D    |    E    |    F    |    G    |    H
    7  Service                 123     9.7%
    8  Total Sales             1273   100.0%
    9
   10  COST OF SALES
   11   Product                 566    91.9%
   12   Service                  50     8.1%
   13  Total C.O.S.             616   100.0%
   14
   15  GROSS MARGIN
   16   Product                 584    50.8%
   17   Service                  73    59.3%
   18  Total G.M.               657    51.6%
   19
   20  EXPENSES
   21    Sales                  255    20.0%
   22  Marketing                140    11.0%
   23  Administrative           170    13.4%
   24  Total Expenses           565    44.4%
   25  OPERATING INCOME          92     7.2%
Formula: =C25/C8
Format: %1  0.0%                      Value: 7.2270227808E-2
Alignment: General      F/B Color: 0/0

Cell A|D25                                <DRI>     11-Jun-91
```

Figure 4-5
Your Worksheet After the **FORMAT** Command

You can now set the format for a group of cells. In the next step, you use the **FORMAT** command again.

WHAT DO I DO IF?

I don't go to the Editing Menu when I press [MENU]?

If you press [MENU] while you're in editing mode, it takes you out of editing mode, but you stay at your worksheet. Press [MENU] again to go to the Editing Menu.

I see the message "The item you selected is not recognized - please try again"?

AlphaCALC doesn't understand what you typed on the command line. Press a key to clear the message, then retype the **FORMAT** command.

Nothing has changed when I return to my worksheet?

If you don't press [EXECUTE] before you return to your worksheet display, the changes you make in **FORMAT** don't take effect. Repeat the [FORMAT] command, being sure to press [EXECUTE] after you change the fields you want.

I see the message "Illegal cell range"?

You typed the cell range incorrectly. Retype it.

My terminal beeps, and nothing happens?

Your entry is not correct for the field. For example, you may have entered L in the format field. AlphaCALC beeps to let you know there is an error, and the cursor stays at the field. Type the correct option and continue.

6 SETTING CELL ALIGNMENT

Looking at your worksheet display, or the copy you printed in step 3, you may notice the titles in column A don't look very professional. They are all different lengths, and don't line up at all. In this step, you align these titles. You also learn an alternative way to use any Editing Menu command.

YOU SEE: Your worksheet display.

TYPE: **/**

YOU SEE: A window listing several options appears over your worksheet, as shown in Figure 4-6.

```
        |    A    |    B    |    C    |    D    |    E    |    F    |    G    |    H
    7    Service                 123      9.7%
    8   Total Sales             1273    100.0%
    9
   10   COST OF SALES
   11    Product                 566     91.9%
   12    Service                  50      8.1%
   13   Total C.O.S.             616    100.0%
   14
   15   GROSS MARGIN
   16    Product                 584     50.8%
   17    Service                  73     59.3%
   18   Total G.M.               657     51.6%
   19
   20   EXPENSES
   21     Sales                  255     20.0%
   22   Marketing                140     11.0%
   23   Administrative           170     13.4%
   24   Total Expenses           565     44.4%
   25   OPERATING INCOME          92      7.2%
+ Worksheet Command Menu -----------------------------------------------------+
!    File       Print        Quit       Range        Worksheet       Loan      !
+------------------------------------------------------------------------------+

Cell A|D25                                  <DRI>    11-Jun-91
```

Figure 4-6
The Worksheet with the Worksheet Command Menu

This is the Worksheet Command Menu. This menu is an alternative way to perform any Editing Menu command. The choices you see are categories of commands (except for `Quit` and `Loan`, which are commands). **FORMAT** is a `Range` command.

TYPE: **R**

YOU SEE: The Worksheet Command Menu is replaced by a window showing the available Range commands.

Notice that you don't need to press [RETURN] or [EXECUTE] when selecting from a Slash Command Menu. Just type the first letter of your choice.

TYPE: **F**

YOU SEE: The Cell Format Information screen.

This screen is exactly the same as the one you used in the last step. Only the method you used to get here is different.

When you are more familiar with slash commands, you don't need to wait for the command menus to appear. You can just type **/RF** at your worksheet display to go directly to the Cell Format Information screen.

TYPE: A [RETURN]

YOU SEE: `(Col)` appears next to the `Cell Range` field, and 9 appears in the `Column width` field, as shown in Figure 4-7.

```
      |    A    |    B    |    C    |    D    |    E    |    F    |    G    |    H
    7  Service                 123     9.7%
    8  Total Sales             1273   100.0%
    9
   10  COST OF SALES
   11   Product                 566    91.9%
   12 +-------------------  Cell Format Information  --------------------+
   13 !                                                                  !
   14 !   Cell Range: A                          (Col)                   !
   15 !       Format: GEN General                                        !
   16 !    Alignment: General                                            !
   17 !   Clear ACUM: No                                                 !
   18 !    F/B Color: 0/0                                                !
   19 ! Column width: 9                                                  !
   20 !                                                                  !
   21 !                                                                  !
   22 !  [ Press EXECUTE to perform set function -OR- MENU to cancel ]   !
   23 +----------------------------------------------------------------+
   24  Total Expenses          565    44.4%
   25  OPERATING INCOME         92     7.2%
Formula: =C25/C8
Format: %1  0.0%                      Value: 7.2270227808E-2
Alignment: General      F/B Color: 0/0
 Format types: GEN; I; IC; N1; N2; N3; N4; N2C; $C; $2; $2C; %; %1; %2; %3; E2
Cell A|D25                                 <DRI>     11-Jun-91
```

Figure 4-7
The Cell Format Information Screen for a Column

By entering just the column letter, you specify the entire column. Since you are changing all the cells in the column, AlphaCALC gives you the option to change the width of the column if you want.

When you enter an entire column—or entire row—in **FORMAT**, you affect ***all*** the cells in it, including empty ones. This means any information you add in column A will use these new settings. However, when you enter a cell range as you did in the last step, you affect only cells containing data.

MOVE: The cursor to the alignment field.

So far, all the cells you've entered have used the general alignment. This is the usual default alignment. It right justifies numbers and centers text.

TYPE: **L**

YOU SEE: The alignment field displays `Left`.

PRESS: EXECUTE

YOU SEE: The cursor moves back to `Cell Range`. The `(Col)` message disappears.

The left justified setting aligns all the titles in column A, starting at the left edge of the cells.

Pressing [EXECUTE] made the new alignment take effect. You could now change the format for another group of cells. Since all the numbers in the worksheet are already aligned, you don't need to change anything else now.

PRESS: [MENU]

YOU SEE: Your worksheet display. This is another difference between using the Editing Menu and using a slash command. After a slash command, you return to your worksheet display, not the Editing Menu. The worksheet now looks like Figure 4-8.

```
      |    A    |    B    |    C    |    D    |    E    |    F    |    G    |    H
    7  Service                 123     9.7%
    8  Total Sales            1273   100.0%
    9
   10  COST OF SALES
   11  Product                 566    91.9%
   12  Service                  50     8.1%
   13  Total C.O.S.            616   100.0%
   14
   15  GROSS MARGIN
   16  Product                 584    50.8%
   17  Service                  73    59.3%
   18  Total G.M.              657    51.6%
   19
   20  EXPENSES
   21  Sales                   255    20.0%
   22  Marketing               140    11.0%
   23  Administrative          170    13.4%
   24  Total Expenses          565    44.4%
   25  OPERATING INCOME         92     7.2%
Formula: =C25/C8
Format: %1  0.0%                      Value: 7.2270227808E-2
Alignment: General      F/B Color: 0/0

Cell A|D25                                  <DRI>    11-Jun-91
```

Figure 4-8
Your Worksheet After Alignment

You probably agree this is a much more professional looking worksheet. You improve its appearance even more in Lesson 5.

WHAT DO I DO IF?

In addition to this problem, see the list of possible symptoms at the end of step 5.

My terminal beeps when I try to enter a slash command?

You've made an incorrect entry for the slash command menu you're at. Be sure to type only the first letter of the command you want, and don't press [RETURN] or [EXECUTE].

7 THE IF FUNCTION

This step introduces one of AlphaCALC's more complicated and powerful functions. The IF function is a logical function, using AlphaCALC's ability to compare two values. It lets you choose which of two values or text strings to put in a cell, depending on another value.

In this step, you enter an IF function that lets you see at a glance if Wizmo, Inc. made a profit or lost money for the month.

YOU SEE: Your worksheet display.

PRESS: [HOME]

YOU SEE: The cursor moves to cell A1. Pressing [HOME] always moves the cursor to cell A1 of the current layer.

The IF function goes after the last row of figures, so you want to move to the end of your worksheet. Here's another way to move the cursor around your worksheet.

PRESS: [SHIFT] / [HOME]

If your keyboard has an [END] key, use it instead of [SHIFT] / [HOME].

YOU SEE: The cursor moves to cell A25.

Pressing [SHIFT] / [HOME] moves the cursor to the bottom of the current layer. It stays in the same column—in this case, column A—and goes to the last row in which any cell contains data.

[SHIFT] / [HOME] is often the fastest way to get to the end of your worksheet to add more data.

MOVE: The cursor to cell C26.

TYPE: **`IF(C25>0,"PROFIT","LOSS")`**

PRESS: [EXECUTE]

YOU SEE: PROFIT appears in the cell and in the `Value` field.

If you express this IF function in words, it goes something like: "If the value of C25—the operating income for the month—is more than 0, display PROFIT in the cell; if C25 is not more than 0, display LOSS."

The IF function is far more complex than SUM. As with SUM, you enter the function name, followed by an "argument" enclosed in parentheses. But the argu ment for IF is more complicated than the cell range you entered for SUM. IF's argument has three parts, with commas between them:

- A logical expression, comparing two values. In this case, it's C25>0, which means "C25 is greater than 0." The ">" symbol means "is greater than."
- An expression or constant. In this case, it's the text string PROFIT. Since this is text, you enclose it in quotation marks.
- Another expression or constant, in this case, LOSS.

The first thing IF does is determine if the logical expression is true. In this case, it is. The value of C25 is 92, which is greater than 0.

If the logical expression is true, IF displays the first of the two expressions or constants; if it's false, it displays the second. Since the logical expression is true, PROFIT displays in the cell.

Your worksheet now looks like Figure 4-9.

```
        |    A    |    B    |    C    |    D    |    E    |    F    |    G    |    H
    8   Total Sales            1273   100.0%
    9
   10   COST OF SALES
   11   Product                 566    91.9%
   12   Service                  50     8.1%
   13   Total C.O.S.            616   100.0%
   14
   15   GROSS MARGIN
   16   Product                 584    50.8%
   17   Service                  73    59.3%
   18   Total G.M.              657    51.6%
   19
   20   EXPENSES
   21   Sales                   255    20.0%
   22   Marketing               140    11.0%
   23   Administrative          170    13.4%
   24   Total Expenses          565    44.4%
   25   OPERATING INCOME         92     7.2%
   26                      PROFIT
Formula: =IF(C25>0,"PROFIT","LOSS")
Format: GEN General                    Value: "PROFIT"
Alignment: General      F/B Color: 0/0

Cell A|C26                                          11-Jun-91
```

Figure 4-9
Your Worksheet With the IF Function

IF is very powerful because you can make its argument very precise and complex. For example, you can include other AlphaCALC functions as part of the argument, so whether this logical expression is true depends on several different values from other parts of your worksheet. You can choose among several different things to display in one cell by setting up the right **if** function.

WHAT DO I DO IF?

I see the message "Expression syntax is invalid" after I press [EXECUTE], or the IF function displays as text?

There is a mistake somewhere in the argument for IF. Check your entry, and make sure it matches the one in the lesson exactly. Forgetting one of the commas or quotation marks or adding a blank space are the most common reasons for this message.

8 THE SAVE COMMAND

You've added a lot of information to your worksheet since you used the FINISH command in Lesson 2. Since you may not have stored the worksheet in permanent storage since then, you'd lose all this work if someone rebooted your computer.

In this step, you use AlphaCALC's SAVE command to store your worksheet without leaving it and returning to the Main Menu.

YOU SEE: Your worksheet display.

As with other Editing Menu commands, there are two ways to use SAVE—from the Editing Menu, or with a slash command.

To use SAVE as a slash command:

TYPE: **/FS**

The F selects the File Commands Menu, and S selects the SAVE command.

YOU SEE: A percentage counter appears briefly as your worksheet is written to disk, then you return to your worksheet display.

If you want to use SAVE from the Editing Menu, do this:

PRESS: [MENU]

YOU SEE: The Editing Menu.

SELECT: The SAVE command.

YOU SEE: A percentage counter appears briefly, then you return to the Editing Menu.

The current version of your worksheet is now stored permanently.

It's a good idea to SAVE your worksheet from time to time, as you just did, as a safeguard against losing your work. It's rare for a computer system to go down or reboot without warning, but there's no reason to take even a slight chance. Also, you may accidentally change or erase information you want to keep. If you SAVE your worksheet, you can retrieve your work. Lesson 5 shows another reason to SAVE your worksheet.

WHAT DO I DO IF?

I see the message "The item you selected is not recognized - please try again"?

AlphaCALC doesn't understand what you typed on the command line. Press a key to clear the message, then retype the SAVE command.

My terminal beeps when I try to enter a slash command?

You've made an incorrect entry for the slash command menu you're at. Be sure to type only the first letter of the command you want, and don't press [RETURN] or [EXECUTE].

LESSON FIVE
REARRANGING YOUR WORKSHEET

Being able to rearrange data in your worksheet easily is one of the wonderful things about AlphaCALC. You can change the format of a report in seconds, or copy data to another section rather than entering it all over again. In this, the last lesson in this tutorial, you learn some of the ways AlphaCALC lets you copy, move, erase, insert and delete information.

1 INSERTING AND DELETING ROWS

From time to time, you need to add or delete rows or columns from your worksheet. You may need to add more information, or take out data which no longer applies. In this step, you learn how to insert and delete rows in your worksheet.

If you left AlphaCALC after Lesson 4, use the CALC command to re-enter it, then the **EDIT** command to return to your INCOME worksheet.

YOU SEE: Your worksheet display.

TURN OFF: Auto-advance mode if it is on.

MOVE: The cursor to cell B2.

PRESS: [ROW INSERT]

YOU SEE: Row 2, and all the rows below it, move down one row. Rows 1-19 of your worksheet look like Figure 5-1.

```
      |    A    |    B    |    C    |    D    |    E    |    F    |    G    |    H
   1          WIZMO, INC.
   2
   3          May 1991
   4
   5                      MONTH      %
   6  SALES
   7  Product                1150    90.3%
   8  Service                 123     9.7%
   9  Total Sales            1273   100.0%
  10
  11  COST OF SALES
  12  Product                 566    91.9%
  13  Service                  50     8.1%
  14  Total C.O.S.            616   100.0%
  15
  16  GROSS MARGIN
  17  Product                 584    50.8%
  18  Service                  73    59.3%
  19  Total G.M.              657    51.6%
Formula:
Format: GEN General
Alignment: General      F/B Color: 0/0

Cell A|B2                                          11-Jun-91
```

Figure 5-1
Your Worksheet After Inserting a Row

You inserted a blank row above row 2. You now have room to enter a heading noting all amounts are in thousands of dollars.

MOVE: The cursor back to cell B2.

ENTER: **`IN THOUSANDS`**

MOVE: The cursor to any cell in row 9.

PRESS: [ROW INSERT]

YOU SEE: A blank row is inserted above row 9. All the data below it moves down one row.

PRESS: [ROW DELETE]

YOU SEE: Row 9 is deleted. All the data below it moves up one row.

[ROW INSERT] always inserts a row at the current cursor position and moves everything below it down one row. You can have the cursor in any cell in the row.

[ROW DELETE] is the opposite of [ROW INSERT]. It deletes the row the cursor is in.

[ROW DELETE] deletes ***all*** information in the row where the cursor is, including data in cells that aren't on your screen. Be careful not to delete information you want to keep.

Both [ROW DELETE] and [ROW INSERT] affect only the current layer.

PRESS: [ROW INSERT]

YOU SEE: A row is inserted at row 9.

MOVE: The cursor to row 15.

PRESS: [ROW INSERT]

You can use either the [LINE INSERT] key or [ROW INSERT] to insert a row, and [LINE DELETE] instead of [ROW DELETE]. These pairs of keys are interchangeable.

You are adding blank rows to your worksheet to improve its format. Later in this lesson, you add rules—dashed lines indicating totals for each category—to your worksheet.

MOVE: The cursor to row 21.

PRESS: [ROW INSERT]

INSERT: One row between `Administrative` and `Total Expenses` and another between `Total Expenses` and `Operating Income`.

YOU SEE: Rows 14-32 of your worksheet now look like Figure 5-2.

```
      |    A    |    B    |    C    |    D    |    E    |    F    |    G    |    H
   14 Service                   50      8.1%
   15
   16 Total C.O.S.             616    100.0%
   17
   18 GROSS MARGIN
   19 Product                  584     50.8%
   20 Service                   73     59.3%
   21
   22 Total G.M.               657     51.6%
   23
   24 EXPENSES
   25 Sales                    255     20.0%
   26 Marketing                140     11.0%
   27 Administrative           170     13.4%
   28
   29 Total Expenses           565     44.4%
   30
   31 OPERATING INCOME          92      7.2%
   32                     PROFIT
Formula:
Format: GEN General
Alignment: Left         F/B Color: 0/0

Cell A|A32                                          11-Jun-91
```

Figure 5-2
Your Worksheet With Rows Inserted

You can insert or delete rows anywhere in your worksheet. Inserting and deleting columns works exactly the same way, except you use the [COLUMN INSERT] and [COLUMN DELETE] keys.

Be careful when you delete rows and columns so you don't delete information you want to keep. Make sure the cursor is in the correct position before you use either of these keys!

WHAT DO I DO IF?

I accidentally erase a row with data in it?

You have two choices:

- If you stored your worksheet immediately before this lesson, use the Editing Menu **QUIT** command, as described in step 9, to leave your worksheet. This returns the worksheet to how it was when you stored it. You can then start the lesson over.
- If you didn't store your worksheet after Lesson 4, use [ROW INSERT] to insert a blank row in place of the one you erased. Then, re-enter all the data that belongs in the row.

I insert a row in the wrong place?

No problem. Just use [ROW DELETE] to remove the extra row.

2 MARKING A BLOCK OF CELLS

Now that you've inserted blanks above all the totals, you can enter ruled lines in the empty cells. However, entering that many rules would be tedious. Instead, you can enter the rules once, then copy them to the other locations.

In this step, you learn the first part of copying a range of cells. This procedure—marking the cells you want to affect—is one of the most common and useful in AlphaCALC, because you can use it with many AlphaCALC keys and commands.

You also learn about a special cell alignment you can use to improve your worksheet's appearance.

YOU SEE: Your worksheet display.

MOVE: The cursor to cell C9.

ENTER: **\-**

YOU SEE: A dashed line appears in cell C9.

You can use the first character of a text entry to set the cell alignment. The \ character tells AlphaCALC you want to use duplicating alignment. In duplicating alignment. AlphaCALC repeats your entry as many times as necessary to fill the cell.

Other text alignment characters are: " for left justified, ’ for right justified, and ^ for centered. You can also set text alignment in the `Alignment` field, in which case AlphaCALC may add one of these characters to the beginning of your entry.

These alignment characters are type identifiers for text, like the equal sign (=) is the type identifier for an expression.

MOVE: The cursor to cell D9.

ENTER: **\-**

YOU SEE: A dashed line appears in cell D9. Your worksheet now looks like Figure 5-3.

```
     |    A    |    B    |    C    |    D    |    E    |    F    |    G    |    H
   1          WIZMO, INC.
   2          In Thousands
   3          May 1991
   4
   5                       MONTH      %
   6  SALES
   7  Product                 1150    90.3%
   8  Service                  123     9.7%
   9                     ------------------
  10  Total Sales             1273   100.0%
  11
  12  COST OF SALES
  13  Product                  566    91.9%
  14  Service                   50     8.1%
  15
  16  Total C.O.S.             616   100.0%
  17
  18  GROSS MARGIN
  19  Product                  584    50.8%
Formula: \-
Format: GEN General
Alignment: Duplicate    F/B Color: 0/0

Cell A|D9                                          11-Jun-91
```

Figure 5-3
Your Worksheet One More Time

You have the single rules for the first category, sales. Now, you want to copy these cells—C9 and D9—to other locations in your worksheet.

> The first step in copying cells is marking the cells you want to copy.

MOVE: The cursor to cell C9.

PRESS: [BLOCK MARK]

YOU SEE: Cell C9 changes to dim display. The message `A|C9:A|C9` appears at the left center of the bottom of your screen. This part of the bottom line of your display always shows the currently marked block.

MOVE: The cursor to cell D9.

PRESS: [BLOCK MARK]

YOU SEE: Cells C9 and D9 are in dim display. The message at the bottom of your screen says `A|C9:A|D9`.

You mark a block by marking the top left and bottom right corners. In this case, the block is only two cells, but you mark larger blocks the same way.

When you mark a block of cells, it changes to dim display so you can see at a glance which cells are marked. A range or block of cells is always a rectangle.

A "block" and a "range" of cells are the same thing. Generally, when you type the beginning and ending cells to define the range—as you did in the **FORMAT** command—it's called a range, and when you use [BLOCK MARK] to mark the cells, it's called a block.

You now know how to mark a block of cells. This procedure can save you a lot of time as you use AlphaCALC. You can perform almost any operation in AlphaCALC—copying, erasing, printing, and so on—on a marked block of cells.

WHAT DO I DO IF?

I mark the wrong block of cells?

To clear the block marks, press [BLOCK CLEAR], as described in step 4. Then, mark the correct cell range.

3 COPYING A BLOCK OF CELLS

You can now copy the marked single rules to any other location in your worksheet. In this step, you copy it to the other sections of your income statement.

YOU SEE: Your worksheet display, with cells C9 and D9 in dim display.

MOVE: The cursor to cell C15.

PRESS: [BLOCK COPY]

YOU SEE: A single rule displays in cells C15 and D15. The formula line shows `\-`, since that is now the cell formula for C15.

MOVE: The cursor to cell C21.

PRESS: [BLOCK COPY]

MOVE: The cursor to cell C28.

PRESS: [BLOCK COPY]

YOU SEE: The single rule is copied to these locations. Rows 11-29 of your worksheet now look like Figure 5-4.

```
        |    A    |    B    |    C    |    D    |    E    |    F    |    G    |    H
   11
   12   COST OF SALES
   13   Product                  566     91.9%
   14   Service                   50      8.1%
   15                       ------------------
   16   Total C.O.S.             616    100.0%
   17
   18   GROSS MARGIN
   19   Product                  584     50.8%
   20   Service                   73     59.3%
   21                       ------------------
   22   Total G.M.               657     51.6%
   23
   24   EXPENSES
   25   Sales                    255     20.0%
   26   Marketing                140     11.0%
   27   Administrative           170     13.4%
   28                       ------------------
   29   Total Expenses           565     44.4%
Formula:
Format: GEN General
Alignment: General      F/B Color: 0/0

Cell A|C6           A|C9:A|D9                        11-Jun-91
```

Figure 5-4
Your Worksheet After Copying the Rules

When you press [BLOCK COPY], the marked cells are copied to the range starting with the current cell.

> The location of the cursor controls where you copy the marked cells to.

BLOCK COPY copies over any existing data in the area you copy to. Be careful not to copy over information you want to keep.

You can make as many copies of the marked range as you want, by moving the cursor to each location.

The ability to copy a cell range to other locations in your worksheet is a great time saver. You'll use it often as you work with AlphaCALC.

WHAT DO I DO IF?

I get the message "No block marked"?

You tried to use BLOCK COPY without marking the block first. Go back to step 2 and mark the block of cells.

I copy the cell range to the wrong location?

If you copy the cells to the wrong place, but don't copy over any existing information, correct it the same way as when you enter data in the wrong cell. Go into editing mode, erase the cell formula, and press EXECUTE. Or, use this procedure:

1. Press BLOCK CLEAR to clear the existing block marks.
2. Use BLOCK MARK to mark the cells you want to erase.
3. Press BLOCK ERASE to erase all the cells in the marked block.

If you do copy over information, follow this procedure, then re-enter the data.

4 CLEARING BLOCK MARKS

You now have single rules everywhere you need them in your worksheet. Since you don't need any more copies of the cells you marked, you should "unmark" the block. In AlphaCALC, this is called "clearing" the block marks.

When you're done using a marked block, either copying it or using one of AlphaCALC's other features, you should ***always*** clear the block marks. Because many AlphaCALC features work only on the marked block if one is marked, you could accidentally perform an operation on the wrong cell range if you forget to clear the marks.

YOU SEE: Your worksheet display, with the cell range C9:D9 marked.

PRESS: BLOCK CLEAR

YOU SEE: The cells return to normal display, and the `A|C9:A|D9` message clears from the bottom of your screen.

That's all there is to it. Press [BLOCK CLEAR] and any marked cell range is cleared. The cursor can be anywhere in your worksheet. Failing to clear the block marks could lead to inconvenience later on, so be sure to do this after you use a marked block.

WHAT DO I DO IF?

I get the message "No block marked"?

You pressed [BLOCK CLEAR] again after you cleared the block marks. Continue with the next step.

5 MORE COPYING

In this step, you add double rules below each section total to your worksheet. Along the way, you practice marking blocks and copying cell ranges.

YOU SEE: Your worksheet display.

ENTER: **\=** in cells C11 and D11.

MOVE: The cursor to cell C11.

PRESS: [BLOCK MARK]

MOVE: The cursor to cell D11.

PRESS: [BLOCK MARK]

YOU SEE: The range C11:D11 in dim display.

You entered the rules once and marked the cells, so you can now copy them to the other locations.

MOVE: The cursor to cell C17.

PRESS: [BLOCK COPY]

MOVE: The cursor to cell C23.

PRESS: [BLOCK COPY]

MOVE: The cursor to cell C30.

PRESS: [BLOCK COPY]

YOU SEE: Your worksheet now looks like Figure 5-5.

```
      |    A    |    B    |    C    |    D    |    E    |    F    |    G    |    H
   12 COST OF SALES
   13 Product                  566    91.9%
   14 Service                   50     8.1%
   15                    ------------------
   16 Total C.O.S.             616   100.0%
   17                    ==================
   18 GROSS MARGIN
   19 Product                  584    50.8%
   20 Service                   73    59.3%
   21                    ------------------
   22 Total G.M.               657    51.6%
   23                    ==================
   24 EXPENSES
   25 Sales                    255    20.0%
   26 Marketing                140    11.0%
   27 Administrative           170    13.4%
   28                    ------------------
   29 Total Expenses           565    44.4%
   30                    ==================
Formula: \=
Format: GEN General
Alignment: Duplicate    F/B Color: 0/0

Cell A|D30           A|C11:A|D11                    11-Jun-91
```

Figure 5-5
Your Worksheet With Rules

PRESS: [BLOCK CLEAR]

YOU SEE: C11 and D11 return to regular display. The `A|C11:A|D11` message clears from your screen.

Always clear the block marks when you're through with a marked block.

You can manipulate marked blocks of cells in many other ways, too. To name just two, AlphaCALC's **DUPLICATE** command lets you make multiple copies of a marked block in one step, and you can mark the block to affect before you use the **FORMAT** command.

WHAT DO I DO IF?

I mark the wrong block of cells?

To clear the block marks, press [BLOCK CLEAR], as described in step 4. Then, mark the correct cell range.

I get the message "No block marked"?

You tried to use [BLOCK COPY] without marking the block first. Go back to step 2 and mark the block of cells.

I copy the cell range to the wrong location?

If you copy the cells to the wrong place, but don't copy over any existing information, correct it the same way as when you enter data in the wrong cell. Go into editing mode, erase the cell formula, and press [EXECUTE]. Or, use this procedure:

1. Press [BLOCK CLEAR] to clear the existing block marks.
2. Use [BLOCK MARK] to mark the cells you want to erase.
3. Press [BLOCK ERASE] to erase all the cells in the marked block.

If you do copy over information, follow this procedure, then re-enter the data.

6 FORMULA ADJUSTMENT

You've finished the monthly section of this income statement. In this step, you see how easy it is to create the year to date section using [BLOCK COPY].

This step illustrates AlphaCALC's "formula adjustment" feature. When you copy a cell which refers to another cell—such as C10, which adds the values of C7 and C8—AlphaCALC automatically adjusts the cell formula to account for its new location. In this step, you copy column C to column E. When you do, all the cell references change from C to E.

Though we didn't mention it at the time, AlphaCALC also automatically adjusted the cell references when you inserted rows in step 1 of this lesson.

YOU SEE: Your worksheet display.

MOVE: The cursor to cell C5.

PRESS: [BLOCK MARK]

MOVE: The cursor to cell D32.

You can use [SHIFT] / [HOME] to move to the end of your worksheet.

PRESS: [BLOCK MARK]

You marked the entire monthly section of the income statement. To create the year to date portion, you copy these cells, and then enter the correct values. Because you copy them, you don't need to re-enter the math calculations.

MOVE: The cursor to cell E5.

PRESS: [BLOCK COPY]

YOU SEE: Cells C5:D32 are copied to cells E5:F32. Your worksheet looks like Figure 5-6.

```
      |    A    |    B    |    C    |    D    |    E    |    F    |    G    |    H
   1          WIZMO, INC.
   2          In Thousands
   3          May 1991
   4
   5                      MONTH      %      MONTH      %
   6  SALES
   7  Product                 1150    90.3%     1150    90.3%
   8  Service                  123     9.7%      123     9.7%
   9                     ------------------------------------
  10  Total Sales             1273   100.0%     1273   100.0%
  11                     ====================================
  12  COST OF SALES
  13  Product                  566    91.9%      566    91.9%
  14  Service                   50     8.1%       50     8.1%
  15                     ------------------------------------
  16  Total C.O.S.             616   100.0%      616   100.0%
  17                     ====================================
  18  GROSS MARGIN
  19  Product                  584    50.8%      584    50.8%
Formula: MONTH
Format: GEN General
Alignment: General      F/B Color: 0/0

Cell A|E5           A|C5:A|D32                      11-Jun-91
```

Figure 5-6
After Copying the Monthly Figures

PRESS: BLOCK CLEAR

MOVE: The cursor to several of the cells in columns E and F.

YOU SEE: The copied formulas all refer to the proper columns.

AlphaCALC can automatically adjust cell formulas for you. Since there may be times when you don't want to change the cell references, there is also a way to enter cell references so they won't be changed if they are moved or copied. See the section on cell references in the *AlphaCALC Reference Book*.

WHAT DO I DO IF?

Refer to the list of possible problems at the end of step 5.

7 STORING YOUR WORKSHEET

You won't enter the year to date values in this lesson. If you want to complete the income statement by entering these figures, see Appendix A - "For More Practice."

As mentioned in Lesson 4, it's a good idea to SAVE your worksheet occasionally as a safety measure. There's another reason, too.

At times, you may change your worksheet, then decide you don't want to keep the change. For example, you may try a new format for your data and decide you like the old one better. If you SAVE your worksheet before trying a change, you have the option of returning to the previous version.

YOU SEE: Your worksheet display.

Remember, there are two ways you can use the SAVE command. Either:

TYPE: **/FS**

Or:

PRESS: [MENU]

SELECT: **SAVE**

PRESS: [MENU]

YOU SEE: A percentage counter appears briefly as your worksheet is saved on the disk. then you return to the Editing Menu or your worksheet display.

The current version of your worksheet is now stored.

You can now return to the current version of your worksheet if you make a change you don't like, or if you accidentally change or erase information.

WHAT DO I DO IF?

I see the message "The item you selected is not recognized - please try again"?

AlphaCALC doesn't understand what you typed on the command line. Press a key to clear the message, then type the command name again.

My terminal beeps when I try to enter a slash command?

You've made an incorrect entry for the slash command menu you're at. Be sure to type only the first letter of the command you want, and don't press [RETURN] or [EXECUTE].

8 THE GLOBAL COMMAND

You've now performed most of AlphaCALC's basic features—entering and editing data, printing and rearranging your worksheet. In this step, you try a more advanced feature, one that could save you a lot of time as you continue to use AlphaCALC.

AlphaCALC's **GLOBAL** command lets you replace one set of characters—called a "string"—with another one, anywhere it appears in your worksheet. This can be very useful when you're modifying a worksheet you originally used for one purpose to another application, or if you decide to change the titles in a large worksheet. Instead of changing each cell, you just tell AlphaCALC to change one word—or cell ID, or number, or whatever—to another, each time it occurs.

YOU SEE: Your worksheet display.

PRESS: [MENU]

YOU SEE: The Editing Menu.

SELECT: **GLOBAL**

YOU SEE: The Replace/Global Function screen.

You can also use **GLOBAL** by typing **`/WG`** at your worksheet display.

The first field on the command screen lets you enter the cell range you want to affect. Since there is no block marked, the default range is the whole worksheet. You don't need to change this field.

MOVE: The cursor to the `Replace:` field using [RETURN] or [TAB].

TYPE: **`TOTAL`**

MOVE: The cursor to the `With:` field.

TYPE: **`COMPLETE`**

YOU SEE: Your screen now looks like Figure 5-7.

```
                      AlphaCALC 2.0(102)   EDITING MENU

Wednesday, June 12, 1991 10:36 AM       14604 bytes in use,  427588 bytes free.

Editing INCOME     Income Statement for Wizmo, Inc.
      +--------------  AlphaCALC Replace/Global Function  ---------------+
HELP  !                                                                  !eet
FINISH!            Cell Range: A|A1:A|F33           (Range)              !nter
SAVE  !                                                                  !ance
QUIT  !   Replace: TOTAL                                                 !ile
      !                                                                  !
SEARCH!      With: COMPLETE_____________________________________         !heets
REPLAC!                                                                  !
GLOBAL!                      Replacing: Formulas                         !
      !                                                                  !
HEADER!                Is this a GLOBAL replacement? Yes                 !
DUPLIC!                                                                  !o file
PROTEC! [ Press EXECUTE key to perform replacement -OR- MENU to cancel ] !t
FORMAT+------------------------------------------------------------------+
MODE      - update CALC.INI commands     LOAN      - loan amortization


Enter command name and press RETURN: GLO
                          Type the replacement text
Cell A|E5                                            11-Jun-91
```

Figure 5-7
The Global Replace Command Screen

You told AlphaCALC to replace "Total" with "Complete" in all the cells of your worksheet. For now, you don't need to worry about the other fields on this screen.

PRESS: [EXECUTE]

YOU SEE: A message at the bottom of your screen tells you **GLOBAL** replaced four instances of "Total" with "Complete."

PRESS: [MENU]

YOU SEE: You return to the Editing Menu.

If you used the /WG slash command to call up **GLOBAL**, you return to your worksheet display. You don't need to press [MENU] again.

PRESS: [MENU]

YOU SEE: Your worksheet display, with the word `Complete` in cells A10, A16, A22 and A30, replacing `Total`. `Total` appeared four times in your worksheet.

You can use **GLOBAL** to change any kind of cell formula—text, as you just did, numbers, cell references, or complicated math formulas. No matter how large your worksheet is, you can change all the occurrences of a string with one command.

When you want to replace only some occurrences of a string, you can use AlphaCALC's **SEARCH** or **REPLACE** command instead of **GLOBAL**. You can use any of these commands on your whole worksheet or any cell range you choose.

WHAT DO I DO IF?

I see the message "Specified string was not found."

One of these things has happened:

- You may have incorrectly typed the string to search for. AlphaCALC finds only exact matches to what you typed. For example, you could have typed "totla" by accident. If you made a mistake, retype the string.
- If you type the string correctly, but still get this message, check the cell range field. If there's a marked block of cells in your worksheet, **GLOBAL** uses it as the default range, or you may have made an entry here by mistake. In either case, make sure this entry is big enough to include all your data (for example `A|A1:A|F32`). Then, press [EXECUTE] again.
- Make sure the `Replacing:` field is still set to "Formulas," not "Expressions."

I see the message "The item you selected is not recognized - please try again"?

AlphaCALC doesn't understand what you typed on the command line. Press a key to clear the message, then type the command name again.

My terminal beeps when I try to enter a slash command?

You've made an incorrect entry for the slash command menu you're at. Be sure to type only the first letter of the command you want, and don't press [RETURN] or [EXECUTE].

9 THE EDITING MENU QUIT COMMAND

Looking at the results, you decide you like the worksheet better with "Total" instead of "Complete" in the row titles. Since you stored your worksheet with the **SAVE** command just before making this change, it's easy to change your worksheet back to the way it was.

The **QUIT** command from the Editing Menu lets you leave your worksheet and return to the Main Menu ***without*** storing your worksheet. When you use Editing Menu **QUIT**, your worksheet returns to the way it was when you last used **FINISH** or **SAVE**. This is useful in two circumstances:

- When you've made changes to your worksheet and decide you don't want to keep them.
- When you just viewed your worksheet, without changing it.

In this step, you use **QUIT** to return your worksheet to how it was before you used **GLOBAL**.

YOU SEE: Your worksheet display.

PRESS: [MENU]

YOU SEE: The Editing Menu.

TYPE: **`QUIT`** [RETURN]

You can **QUIT** directly from your worksheet by typing **`/Q`**.

YOU SEE: The message `Are you SURE you want to leave this worksheet and abandon all your changes?`.

Since you have changed your worksheet since you last stored it, AlphaCALC asks you to verify that you don't want to store those changes. Since the only change was from "Total" to "Complete," you do want to return to the previous version.

Be very careful when using **QUIT**! It erases ***ALL*** changes since the last time you used **SAVE** or **FINISH**. Make sure you don't lose information you want to keep, or you have to re-enter it!

TYPE: **YES** [RETURN]

YOU SEE: The AlphaCALC Main Menu.

You can now go back and look at your worksheet, or print it, to verify the titles have returned to "Total."

The ability to **QUIT** your worksheet makes the **SAVE** command even more useful. Before you make a change you're not sure you'll like, such as the one you made with **GLOBAL**, **SAVE** your worksheet. That way, if you don't like the new appearance, you can **QUIT** and return to the way your worksheet was in a matter of seconds.

WHAT DO I DO IF?

I see the message "The item you selected is not recognized - please try again"?

AlphaCALC doesn't understand what you typed on the command line. Press a key to clear the message, then type the command name again.

My terminal beeps when I try to enter a slash command?

You've made an incorrect entry for the slash command menu you're at. Be sure to type only the first letter of the command you want, and don't press [RETURN] or [EXECUTE].

A LAST WORD

You’ve completed *The AlphaCALC Tutorial*! You now know enough to use AlphaCALC in many everyday applications. More important, you’ve learned enough about the way AlphaCALC operates that—with a little practice—you can figure out how to use the features not included in these lessons.

If you want to practice more with the sample worksheet, see Appendix A - "For More Practice." In it are the year to date figures for the income statement, and other sections showing the planned income for the company and comparing the planned and actual income.

For complete information about most AlphaCALC features, see the *AlphaCALC Reference Book*. Some of the very technical features are covered in the *AlphaCALC Installation and Administrator’s Guide* or *AlphaCALC Programmer’s Information Manual*.

APPENDIX A
FOR MORE PRACTICE

This appendix contains information to complete the income statement worksheet. If you don't feel completely comfortable or confident with AlphaCALC yet, this material gives you more practice entering cell formulas and working with marked blocks of cells.

So far, you've entered the monthly figures for the income statement. In this appendix, you add:

- The year to date information. You created this section in Lesson 5, but did not enter the data.
- The planned amounts for the month and year, assuming the figures you entered in the lessons are the actual amounts.
- A section comparing the planned and actual amounts.

The instructions and screen descriptions in this appendix are not as detailed as in the lessons. If you need more information on any topic, look back at the lesson where it was introduced or look it up in the *AlphaCALC Reference Book*.

1 THE YEAR TO DATE AMOUNTS

When you copied columns C and D to columns E and F during Lesson 5, AlphaCALC adjusted all the cell references in the cell formulas to account for the new location. To complete the year to date section, return to your worksheet and follow these instructions.

MARK: The cell range B1:B3—the statement titles—using [BLOCK MARK].

MOVE: The cursor to cell D1.

PRESS: [BLOCK COPY]

PRESS: [BLOCK ERASE]

[BLOCK ERASE] removes the formulas from the marked cells and clears the block marks.

YOU SEE: The titles move to column D, centered above the data. Using [BLOCK COPY] and [BLOCK ERASE] together lets you move information from one location to another.

CHANGE: Cell E5 to say **`Y.T.D.`**.

ENTER: The yearly figures, as shown here:

Cell	Amount
E7	5600
E8	666
E13	3333
E14	222
E25	500
E26	300
E27	390

You can enter these cell formulas—and all the others in this appendix—with or without using [AUTO ADVANCE]. You may want to try both with and without this feature to see which way you feel most comfortable.

PRESS: [RECOMPUTE]

YOU SEE: The totals and percentages in columns E and F change to reflect the new figures. Rows 13-31 of your worksheet now look like Figure A-1.

```
     |    A    |    B    |    C    |    D    |    E    |    F    |    G    |    H
  13 Product                 566    91.9%     3333    93.8%
  14 Service                  50     8.1%      222     6.2%
  15                    ------------------------------------
  16 Total C.O.S.            616   100.0%     3555   100.0%
  17                    ====================================
  18 GROSS MARGIN
  19 Product                 584    50.8%     2267    40.5%
  20 Service                  73    59.3%      444    66.7%
  21                    ------------------------------------
  22 Total G.M.              657    51.6%     2711    43.3%
  23                    ====================================
  24 EXPENSES
  25 Sales                   255    20.0%      500     8.0%
  26 Marketing               140    10.9%      300     4.8%
  27 Administrative          170    13.3%      390     6.2%
  28                    ------------------------------------
  29 Total Expenses          565    44.3%     1190    19.0%
  30                    ====================================
  31 OPERATING INCOME         92     7.2%     1521    24.3%
Formula:
Format: GEN General
Alignment: Left         F/B Color: 0/0

Cell A|A28                                          11-Jun-91
```

Figure A-1
Your Worksheet with Yearly Figures

That's all there is to entering the yearly amounts. Using [BLOCK COPY] saves you the effort of entering the formulas again.

2 THE PLANNED AMOUNTS

Your worksheet now includes the amounts for both the month and year. However, are these figures better or worse than Wizmo, Inc. expected? Does an operating income of $92,000 make May a successful month? To figure this out, you need to compare the actual amounts with the planned amounts for the same period.

The procedure for adding the planned amounts to your worksheet is essentially the same as adding the year to date figures. You copy a section of your worksheet—in this case, the entire thing—enter the new amounts, and press [RECOMPUTE] to calculate the totals and percentages.

This is a very common way to build a worksheet. In general, if you can copy and modify an existing section of your worksheet to create a new section, do so. You almost always save work.

To create the "Plan" section of your worksheet, follow these steps.

INSERT: A row at row 4.

ENTER: In cell D4, **ACTUAL**.

This lets you distinguish between sections of the income statement.

MARK: All of your existing worksheet, cells A1:F33, using [BLOCK MARK].

MOVE: The cursor to cell A40.

PRESS: [BLOCK COPY]

Remember to use [BLOCK CLEAR] after you copy the cells, or it could cause problems later.

CHANGE: Cell D43 to say **PLAN** instead of ACTUAL.

ENTER: The planned amounts, as shown:

Cell	Amount	Cell	Amount
C47	1111	E47	4444
C48	111	E48	888
C53	550	E53	2250
C54	42	E54	333
C65	200	E65	550
C66	150	E66	310
C67	180	E67	420

PRESS: [RECOMPUTE]

YOU SEE: The totals and percentages for the planned section.

Rows 46-72 now look like Figure A-2. This figure shows more than the nineteen rows most terminals display, and doesn't include the cell editing fields at the bottom of your worksheet display.

```
    |    A    |    B    |    C    |    D    |    E    |    F    |    G    |    H
 46 SALES
 47 Product                 1111     90.9%      4444     83.3%
 48 Service                  111      9.1%       888     16.7%
 49                    ------------------------------------
 50 Total Sales             1222    100.0%      5332    100.0%
 51                    ====================================
 52 COST OF SALES
 53 Product                  550     92.9%      2250     87.1%
 54 Service                   42      7.1%       333     12.9%
 55                    ------------------------------------
 56 Total C.O.S.             592    100.0%      2583    100.0%
 57                    ====================================
 58 GROSS MARGIN
 59 Product                  561     50.5%      2194     49.4%
 60 Service                   69     62.2%       555     62.5%
 61                    ------------------------------------
 62 Total G.M.               630     51.6%      2749     51.6%
 63                    ====================================
 64 EXPENSES
 65 Sales                    200     16.4%       550     10.3%
 66 Marketing                150     12.3%       310      5.8%
 67 Administrative           180     14.7%       420      7.9%
 68                    ------------------------------------
 69 Total Expenses           530     43.4%      1280     24.0%
 70                    ====================================
 71 OPERATING INCOME         100      8.2%      1469     27.6%
 72                     PROFIT             PROFIT
```

Figure A-2
The Planned Income Statement

3 COMPARING PLANNED AND ACTUAL

Creating the comparison of planned and actual amounts is a little more complicated version of the procedure you used in step 2. Again, you create the framework of the section by copying an existing cell range. But, instead of just changing the numbers, you need to enter math formulas comparing the amounts in the two existing sections.

You can copy either the actual (cell A1:F33) or planned (cells A40:F72) section of your worksheet to create the comparison.

MARK: The section you want to copy, using [BLOCK MARK].

MOVE: The cursor to cell A80.

PRESS: [BLOCK COPY]

CHANGE: Cell D83 to say **`ACTUAL VS. PLAN`**.

This section shows the difference between the planned and actual figures for the company. If the difference increases Wizmo's profits—if the sales and income are higher or expenses and costs are lower—the amount shown is positive. If the difference lowers profits—lower sales or higher costs—the amount is negative. The percentages shown are the percentage differences from the planned amount.

If you've never worked with an income statement, this section may not mean very much to you. You can either omit this part, or go ahead and enter the formulas for the practice.

ENTER: The formulas below. Formulas not listed here stay the same.

Some of these math formulas are more complicated than anything you've entered so far, using parentheses to determine the order of operations. The operation inside the parentheses is always performed first. Be careful to enter the formulas exactly as shown.

Cell	Amount	Cell	Amount
C87	C8-C47	D90	C90/C50
C88	C9-C48	D93	C93/C53
C93	C53-C14	D94	C94/C54
C94	C54-C15	D96	C96/C56
C99	C87+C93	D99	(C20-C59)/C59
C100	C88+C94	D100	(C21-C60)/C60
C105	C65-C26	D105	C105/C65
C106	C66-C27	D106	C106/C66
C107	C67-C28	D107	C107/C67
D87	C87/C47	D109	C109/C69
D88	C88/C48	D111	C111/C71

PRESS: [RECOMPUTE]

YOU SEE: The differences in the monthly amounts, as Figure A-3 shows:

```
     |    A    |    B    |    C    |    D    |    E    |    F    |    G    |    H
 85                          MONTH      %     Y.T.D.      %
 86  SALES
 87  Product                    39     3.5%     4444    83.3%
 88  Service                    12    10.8%      888    16.7%
 89                      ------------------------------------
 90  Total Sales                51     4.2%     5332   100.0%
 91                      ====================================
 92  COST OF SALES
 93  Product                   -16    -2.9%     2250    87.1%
 94  Service                    -8   -19.0%      333    12.9%
 95                      ------------------------------------
 96  Total C.O.S.              -24    -4.1%     2583   100.0%
 97                      ====================================
 98  GROSS MARGIN
 99  Product                    23     4.1%     2194    49.4%
100  Service                     4     5.8%      555    62.5%
101                      ------------------------------------
102  Total G.M.                 27    52.9%     2749    51.6%
103                      ====================================
104  EXPENSES
105  Sales                     -55   -27.6%      550    10.3%
106  Marketing                  10     6.7%      310     5.8%
107  Administrative             10     5.6%      420     7.9%
108                      ------------------------------------
109  Total Expenses            -35    -6.6%     1280    24.0%
110                      ====================================
111  OPERATING INCOME           62    62.0%     1469    27.6%
```

Figure A-3
After Entering Monthly Planned vs. Actual

Now for the year to date figures. Once again, [BLOCK COPY] saves a great deal of work. Be sure to clear any block marks before you follow these instructions.

MARK: Cells C87:D112 using [BLOCK MARK].

COPY: The marked cells to E87:F112 using [BLOCK COPY].

PRESS: [RECOMPUTE]

YOU SEE: Rows 87-112 of your worksheet now look like Figure A-4.

	A	B	C	D	E	F	G	H
87	Product		39	3.5%	1156	26.0%		
88	Service		12	10.8%	-222	-25.0%		
89								
90	Total Sales		51	4.2%	934	17.5%		
91								
92	COST OF SALES							
93	Product		-16	-2.9%	-1083	-48.1%		
94	Service		-8	-19.0%	111	33.3%		
95								
96	Total C.O.S.		-24	-4.1%	-972	-37.6%		
97								
98	GROSS MARGIN							
99	Product		23	4.1%	73	3.3%		
100	Service		4	5.8%	-111	-20.0%		
101								
102	Total G.M.		27	52.9%	-38	-4.1%		
103								
104	EXPENSES							
105	Sales		-55	-27.5%	50	9.1%		
106	Marketing		10	6.7%	10	3.2%		
107	Administrative		10	5.6%	30	7.1%		
108								
109	Total Expenses		-35	-6.6%	90	7.0%		
110								
111	OPERATING INCOME		62	62.0%	-128	-8.7%		
112		PROFIT			LOSS			

Figure A-4
Actual vs Planned Figures

Instead of having to enter 22 formulas, you calculated the year to date amounts in a matter of seconds.

Your income statement is finished. It is now much longer than one printed page, so if you want to print it, you may want to use the Worksheet Header and/or the page break feature to set where the pages will be divided before you do. See the sections on these commands, and the **PRINT** command discussion on determining page size, in the *AlphaCALC Reference Book*.

Now, go ahead and use AlphaCALC in any way you want. Feel free to experiment with different commands and ways of working in your worksheet.

For more information on any AlphaCALC feature discussed in this tutorial, and to learn about the AlphaCALC features not mentioned here (there are lots of them), see the *AlphaCALC Reference Book*.

GLOSSARY
TERMS USED IN THIS TUTORIAL

This glossary describes terms in this tutorial you may not be familiar with. This includes words and phrases particular to AlphaCALC or Alpha Micro—such as AMOS—and some general computer terminology.

Account

A way of organizing files in your computer's disk storage. An account usually contains files related to each other in some way. Generally, you must be logged into an account to use the files stored in the account. Accounts are sometimes called disk accounts.

Account Number

The number identifying an account, which you use when logging on to your computer. Alpha Micro account numbers consist of two parts, a project number and a programmer number, separated by a comma. For example, 7,13 or 1,2. Account numbers are sometimes called Project-programmer numbers, or PPNs.

AMOS

The Alpha Micro Operating System. The operating system software enables your computer to understand your instructions and use software packages such as AlphaCALC.

AMOS Command Level

When you are at AMOS command level, you communicate directly with AMOS. Often, you are at AMOS command level after logging on to your computer. From AMOS command level, you can enter any AMOS command, or call up a software package such as AlphaCALC.

AMOS Prompt

The prompt you see when you are at AMOS command level. This is often a period (.), but may be set to something else.

Argument

The part of a cell formula in parentheses following a function name. It specifies the information for the function to work with.

Arithmetic Operator

A symbol used to perform an arithmetic operation with two values. AlphaCALC has arithmetic operators for addition, subtraction, multiplication, division, and others.

Block of Cells

A group of cells you have marked to use with an AlphaCALC feature. A block of cells is a rectangular section of one or more layers.

Cell

The basic unit of a worksheet, formed by the intersection of a row and a column on a particular layer. Each cell can contain one cell formula. A cell is identified by its cell ID.

Cell Contents

See Cell Value.

Cell Formula

The entry you make in a cell. There are two types of cell formulas: constants, which are either numbers of text; and expressions. AlphaCALC evaluates the cell formula to figure the cell value.

Cell ID

A combination of letters and numbers identifying a particular cell. A complete cell ID contains one or two letters for the layer, the split bar character (|), one or two letters for the column, and a number for the row. For example, A|A1 or AB|J42. When you’re referring to a cell on the current layer, you only need to include the column and row. For example, A1 or AF152.

Cell Range

A group of cells you can use as part of a cell formula or with many AlphaCALC commands. A cell range is a rectangular section of one or more layers. When you mark a cell range using the [BLOCK MARK] key, it is also called a block of cells.

On some command screens, such as **FORMAT**, you can enter a layer, column or row identification in the cell range field.

Cell Value

The value of a cell formula. For constants, this is the same as the cell formula. For expressions, it is the value of the calculated expression. For expressions, the cell value displays in the `Value` field.

There are three types of cell values: numeric, logical (true or false), and text.

Command

An order you type at AMOS command level, or select at one of AlphaCALC's menus, instructing your computer to perform a particular action.

Command Line

The prompt `Enter command name and press RETURN:` at the bottom of the AlphaCALC Main Menu and Editing Menu, where you can type the name of the command you want.

Command Screen

The screen display that appears after you select a command, where you enter information and choose options for the command.

Constant

A cell formula where the value displayed in the cell is the same as your entry (allowing for differences due to cell format). Constants can be either numbers or text. See also Expression.

When a cell formula is a constant, the `Value` field is blank.

Current Cell

The cell currently highlighted by the cursor in your worksheet. Any entry you make is placed in the current cell.

Cursor

A lighted block or underline showing where on the screen your next keyboard entry will appear.

There are two kinds of cursor in AlphaCALC: on menus and command screens and in the cell editing fields, a cursor shows where the next character you type will appear; in your worksheet display, a cursor the size of one cell highlights the current cell.

Default

An entry or response which appears in a field automatically. You can either accept the default response or type another entry.

Directory

A list of the AlphaCALC worksheets in this account. You can see this using the **DIRECTORY** command, or any time a command asks you for a worksheet name.

Editing Mode

The mode in which you enter or change a cell formula. You enter editing mode by starting to type a cell formula, or by pressing the [EDIT] key. While you are in editing mode, you cannot move to a different cell.

Expression

A cell formula where the cell value does ***not*** equal the cell formula. For example, "2+2" is an expression, because the cell formula is "2+2" and the cell value is 4. An expression can be either numeric, logical, or string, depending on its value. Also, see Constant.

When a cell formula is an expression, the value of the expression appears in the `Value` field.

Field

An area of your screen where you fill in information, such as the Description and Author fields in the Worksheet Header.

File

A collection of related information stored together on your computer. For example, an AlphaCALC worksheet is a file.

Formula Adjustment

When a cell formula refers to another cell, and you copy the cell formula to another cell, the cell reference changes to account for the new location. Though it is not discussed in this tutorial, there is a way you can prevent formula adjustment from occurring. See the *AlphaCALC Reference Book*.

Formula adjustment can also occur when you insert or delete rows or columns in your worksheet.

Function

A word or abbreviation you can enter as part of a cell formula. AlphaCALC understands the function to mean "perform this operation." For example, SUM tells AlphaCALC to add a range of cell values. See Argument.

Part 3 of the *AlphaCALC Reference Book* lists and explains AlphaCALC's functions.

Function Keys

One of a group of keys on your keyboard, usually labeled with F and a number, such as F3. You can use function keys to perform many worksheet editing features in Alpha-CALC. For most terminals, you can get a keyboard overlay showing which function key performs each feature.

Logical Expression

Any expression with a logical value: either true or false. Logical expressions normally compare two or more values using one or more logical operators and/or logical functions.

Logical Operator

A symbol or word used to compare two values in an expression. The logical operators available in AlphaCALC are listed in the *AlphaCALC Reference Book*.

Menu

A screen display showing a list of commands for you to choose from. AlphaCALC's menus include the Main Menu, Editing Menu, and the worksheet command menus.

Numeric Expression

Any expression with a numeric value. See Expression.

Printer Queue

A list of files waiting to print or currently printing on a particular printer. The printer queue is maintained by the operating system, AMOS. You can view or modify the printer queue using the **QUEUE** command.

Prompt

A question or request for information on a command screen. You type your response in the field following the prompt.

Slash Command

A command you choose from your worksheet by typing a slash (/) and one or two letters. Except for the page break feature, slash commands are duplicates of the Editing Menu commands.

String

A consecutive group of characters, such as "text." Also called a text string.

String Expression

Any expression with a string value. See Expression.

Type Identifier

A character at the beginning of a cell formula which identifies it as a particular type of formula. For example, the equal sign (=) marks a formula as an expression. If you do not enter a type identifier, AlphaCALC may add it to the formula.

Universal Key Sequence

A sequence of keys, involving either the [CTRL] or [ESCAPE] key, or both, followed by a character. You can use the universal key sequence instead of a function or special key to invoke an AlphaCALC feature on any terminal AlphaCALC supports.

Worksheet

A type of file containing information in a special format, created and understood by AlphaCALC. A worksheet consists of individual cells arranged in rows, columns and layers.

Worksheet Display

The part of your worksheet shown on your terminal screen while you are viewing or editing your worksheet.

DOCUMENT HISTORY

REVISION	DATE	DESCRIPTION
00	10/3/91	AlphaCALC 2.0 release. Replaces previous tutorial, DSO-00008-00. Update whole book with correct 2.0 procedures. Add multiple layer information, etc.

INDEX

A

Addition 2-13
  using the SUM function 2-17
Alignment — see Cell alignment
AlphaCALC's design I-2
AMOS command level 1-1, 1-13, 2-1
Arguments
  for IF function 4-18
  for SUM function 2-17
Arrow keys
  at your worksheet display 2-4
Audience
  intended I-3
Auto advance key 2-11

B

Block clear key 5-8, 5-9
Block copy key 5-6, 5-7, A-2
Block erase key A-1
Block mark key 5-4 to 5-6
Block of cells — see Cell range

C

CALC command 1-2, 2-1
Calling up AlphaCALC 1-2, 2-1
Cell alignment 4-13 to 4-16
  duplicating 5-5
Cell editing fields 1-10
  moving the cursor 4-7
Cell format 4-6, 4-7, 4-9 to 4-11
Cell formulas
  adjusting cell references 5-11, 5-12
  changing 4-4
  constants 2-14
  definition 2-6, 2-14
  entering 2-7
  expressions 2-14
  using parentheses A-5
Cell ID 2-4
  definition I-2
Cell range
  clearing 5-8, 5-9
  copying 5-4 to 5-7, A-2
  definition 5-6
  erasing A-1
  marking 5-4 to 5-6
  moving A-1
  setting cell alignment 4-13 to 4-16
  setting cell format 4-9 to 4-11
Cell references
  benefits 4-6
  formula adjustment 5-11, 5-12
Cell values
  definition 2-14
  recalculating 4-5
Changing a cell formula 4-4
Character delete key
  in editing mode 4-4
Character insert key
  in editing mode 4-4
Clearing block marks 5-8, 5-9
Column delete key 5-4
Column insert key 5-4
Columns
  using with the format command 4-14
Command screens
  correcting mistakes 1-7
  moving around 1-7
Commands
  choosing 1-2, 1-12, 2-1, 4-19
Constant
  definition 2-14
Conventions used in this book I-5
Copying a range of cells 5-4 to 5-7
  formula adjustment 5-11, 5-12
Correcting mistakes
  in cell formulas 4-4
  on command screens 1-7
  while entering commands 1-3
Create command 1-4
Creating a worksheet 1-4
Current cell 1-10
Cursor
  moving around your worksheet 2-4

D

Data
  entering in a cell 2-7
Default entries
  definition 1-6
DEFWKS worksheet 1-5
Deleting columns 5-4
Deleting rows 5-1 to 5-3
Description
  in worksheet header 1-6
Design of AlphaCALC I-2
Directory
  viewing during print command 3-2
Duplicating alignment 5-5

E

Edit command 2-1, 2-2
Edit key 4-4
Editing Menu
  illustration 1-11
  returning to your worksheet 1-11
Editing mode 2-8
  making corrections in 4-4
  moving among fields 4-7
End key 4-17
Entering cell formulas 2-7
Entering text 2-9
Equal sign (=)
  as type identifier 2-14
Erasing a cell range A-1
Execute key
  entering cell formulas 2-7
  on command screens 1-5
Expression
  definition 2-14

F

Finding and replacing a string of characters 5-13, 5-15
Finish command 1-12
Format, cell — see Cell format
Format command 4-9 to 4-11, 4-13 to 4-16
Format field 4-7
Formula adjustment 5-11, 5-12
Formula field 2-7
Function keys I-7
Functions 2-17

G

Global command 5-13, 5-15
Goto key 2-4
Graphic conventions I-5

H

Help command 1-2
Help information
  from a menu 1-2
  from the worksheet display 2-5
  on cell formulas 2-6
  on creating a worksheet 1-3
  viewing 1-3
Help key 2-5, 2-6
Home key 2-5, 4-17

I

IF function 4-17, 4-18
Illustrations
  possible differences from your display I-7
INCOME worksheet
  description I-8
Information Menu 1-3, 2-5
Inserting columns 5-4
Inserting rows 5-1 to 5-3

K

Keys
  function and special I-7

L

Layer letter
  omitting 2-5
Leaving AlphaCALC 1-13
Leaving the worksheet without storing 5-16, 5-17
Left justification 4-15
Lesson organization I-5
Line delete key 5-3
Line insert key 5-3
List of worksheets
  viewing 3-2
Logical expressions 4-17, 4-18

M

Main Menu
  illustration 1-2
Marking a block of cells 5-4 to 5-6
Menu key
  and Editing Menu commands 4-12
  and your worksheet display 1-9
  from the Editing Menu 1-11
Moving a cell range A-1
Moving around command screens 1-7
Moving around your worksheet 2-4, 4-1, 4-2, 4-17

N

Next screen key 4-2
Numbers
  entering 2-7

O

Organization of the lessons I-5

P

Page format
  of the tutorial I-5
Parentheses
  in cell formulas A-5
Percentages 4-6, 4-7
Previous screen key 4-2
Print command 3-1 to 3-5
Printer
  selecting 3-4
Printing the worksheet 3-1 to 3-5
Protection field
  in worksheet header 1-8
Prototype worksheet 1-5

Q

Quit command
  Editing Menu 5-16, 5-17
  Main Menu 1-13

R

Range of cells — see Cell range
Recalculating worksheet values 4-5
Recompute key 4-5
References
  other AlphaCALC documents 1-1
Row delete key 5-1 to 5-3
Row insert key 5-1 to 5-3
Rubout key
  while entering commands 1-3

S

Sample worksheet
  description I-8
Save command 4-19, 5-12
Screen display
  possible differences from screen illustrations I-7
Screen left key 4-2
Screen right key 4-2
Slash commands 4-13 to 4-16
Special keys I-7
Starting AlphaCALC 1-2
Storing your worksheet 1-12, 4-19, 5-12
String of characters
  finding and replacing 5-13, 5-15
Structure of a worksheet I-1
SUM function 2-17

T

Text
  entering 2-9
  when an entry is longer than the cell 2-12
Text entries
  type identifiers 5-5
Type identifiers
  for expressions 2-14
  for text 5-5

U

Universal key sequences I-7

V

Value field 2-14, 4-7

W

Worksheet
  maximum size I-2
  moving around 2-4, 4-1, 4-2
  storing 1-12
  structure I-1
Worksheet directory 3-2
Worksheet display 1-9, 1-10
  illustration 1-10
  moving 4-1, 4-2, 4-17
  viewing help information 2-5
Worksheet header 1-6, 2-3
Worksheets
  calling up an existing 2-1
  creating 1-4
  printing 3-1 to 3-5
  prototype 1-5